

Supplementary Information for

Co-Deposited Inverted Perovskite Photovoltaics towards 27% Efficiency via Vertical Redistribution of Self-Assembled-Molecules and In-Situ Crosslinking

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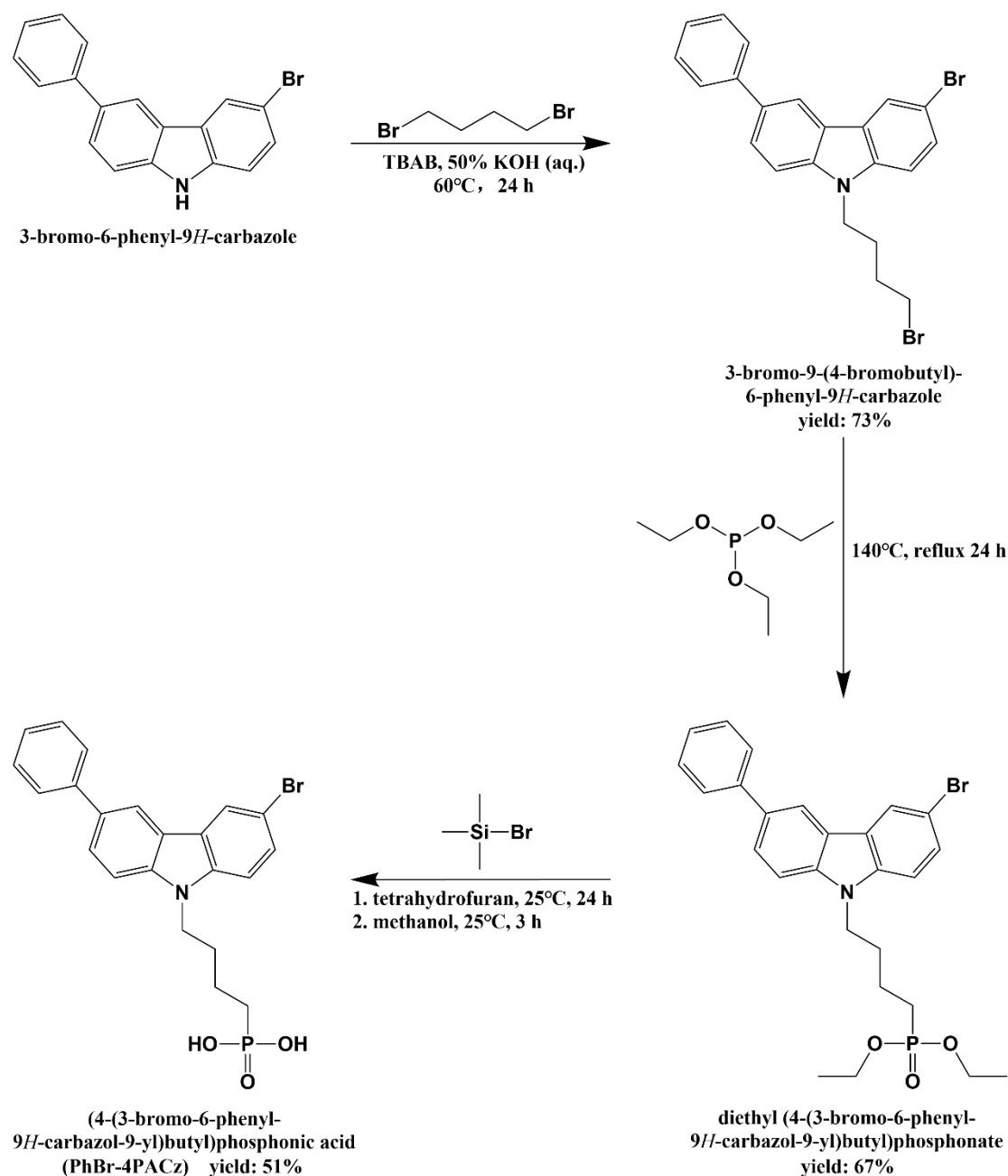
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Supplementary Fig. 1. Synthesis route for PhBr-4PACz.

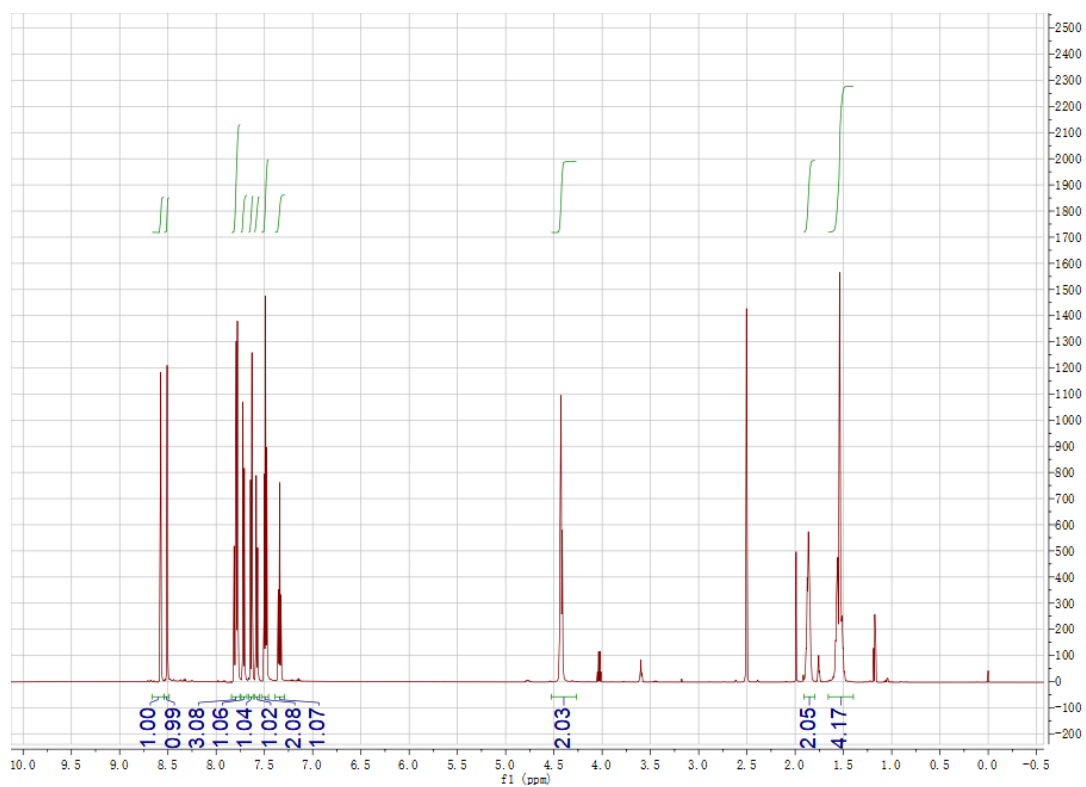
The corresponding detailed synthesis procedures are provided below:

(1) **3-bromo-9-(4-bromobutyl)-6-phenyl-9H-carbazole.** To a solution of 3-bromo-6-phenyl-9H-carbazole (2.00 g, 6.23 mmol), tetrabutylammonium bromide (TBAB, 0.31 g, 0.92 mmol), 1,4-dibromobutane (33.8 g, 156.5 mmol) was added 50% KOH aqueous solution (6.99 g, 62.3 mmol). The mixture was heated to 60°C and stirred for 24 h. After cooling to room temperature, extraction was done with dichloromethane. The organic layer was dried over anhydrous MgSO₄ and filtered. After removal of the solvent under reduced pressure, the crude product was purified by column chromatography (dichloromethane: petroleum ether, 1:5 v/v) to give 2.07 g of colorless oily liquid (73% yield).

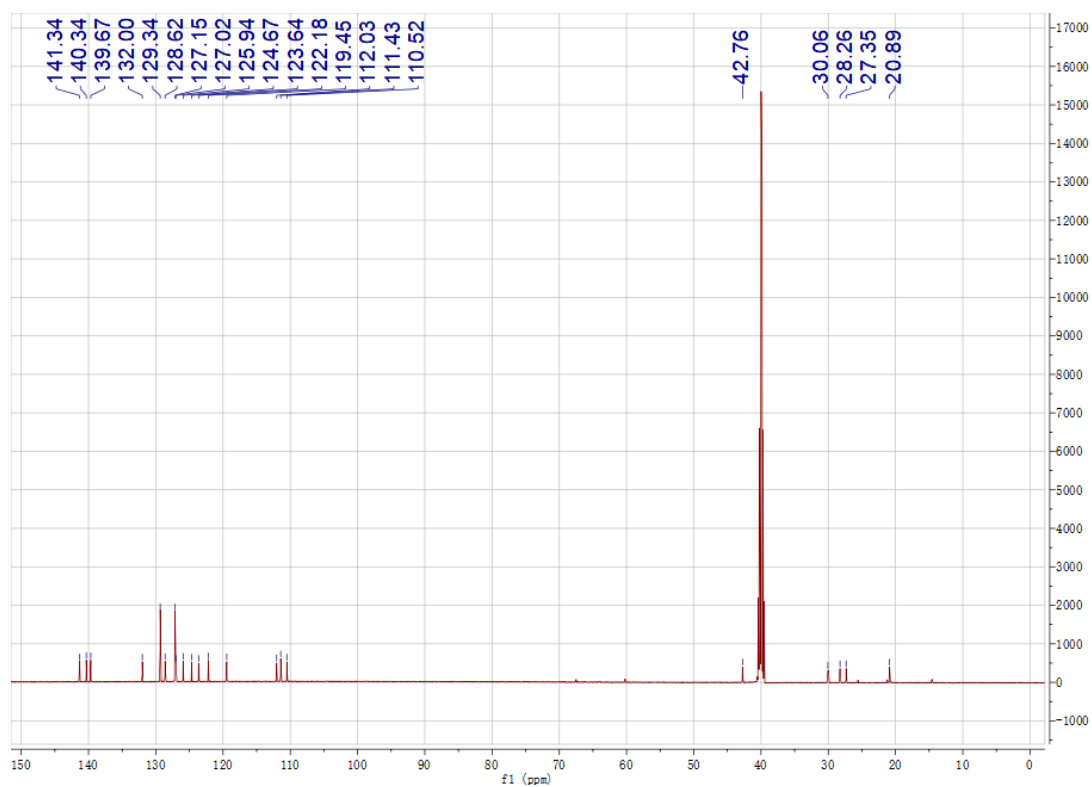
(2) **diethyl (4-(3-bromo-6-phenyl-9H-carbazol-9-yl)butyl)phosphonate.** The solution of 3-bromo-9-(4-bromobutyl)-6-phenyl-9H-carbazole (2.00 g, 4.40 mmol) and triethyl phosphite (18.31

g, 110.20 mmol) was heated to 140°C under reflux and stirred for 24 h under N₂. Then, solvent was distilled off under reduced pressure after cooling to room temperature, the residue was purified by column chromatography (ethyl acetate: petroleum ether, 1:3 v/v) to give 1.52 g of colorless oily liquid (67% yield).

(3) **PhBr-4PACz**. To a solution of diethyl (4-(3-bromo-6-phenyl-9*H*-carbazol-9-yl)butyl)phosphonate (1.50 g, 2.92 mmol) in anhydrous tetrahydrofuran (20 mL) was added bromotrimethylsilane (4.46 g, 29.2 mmol) dropwise at room temperature and stirred for 24 h under N₂. Then, (30 mL) was added to quench the reaction and stirred for 3 h. After that, deionized water (200 mL) was added and stirred for 24 h, the resulting mixture was filtered to give filter residue. Subsequently, the residue was dissolved in tetrahydrofuran and recrystallized by adding petroleum ether three times to give 0.68 g of white solid (51% yield).

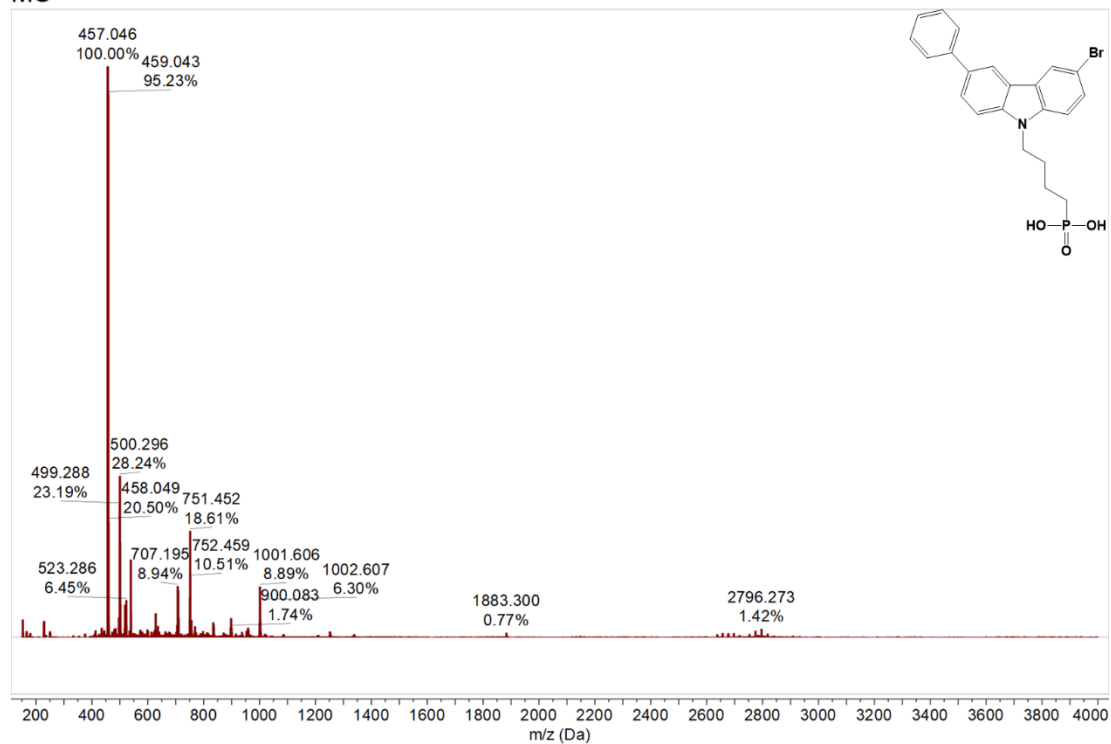


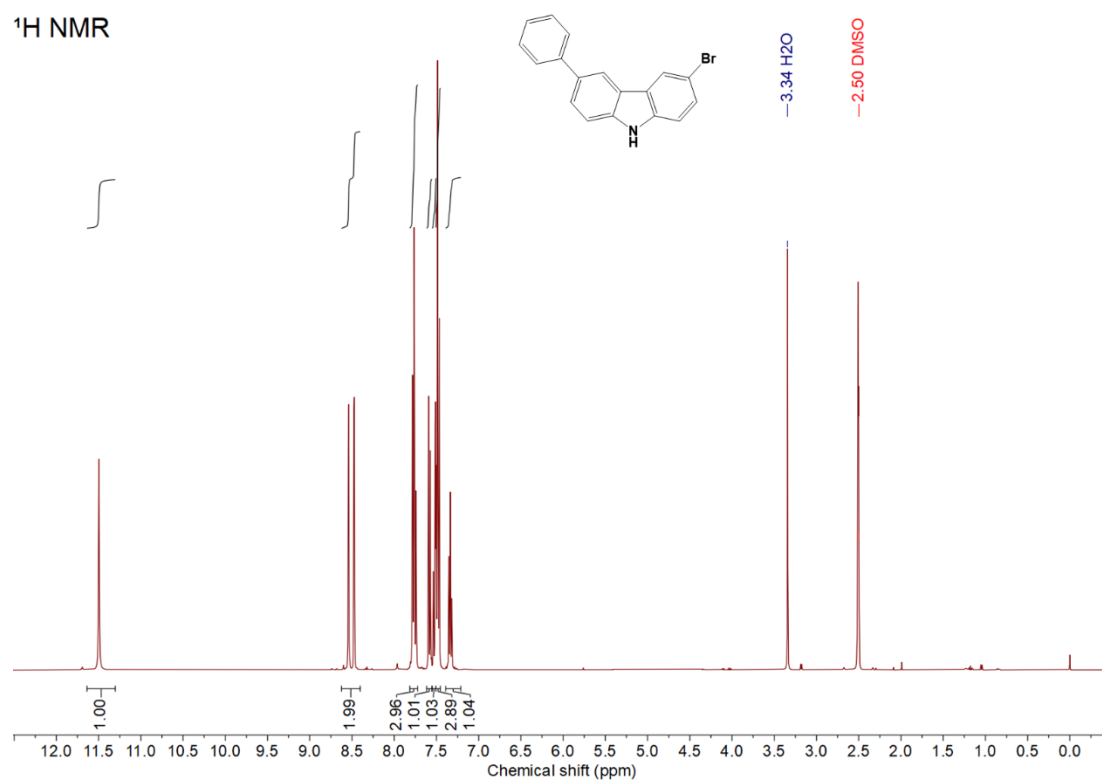
Supplementary Fig. 2. ¹H NMR of PhBr-4PACz in DMSO.



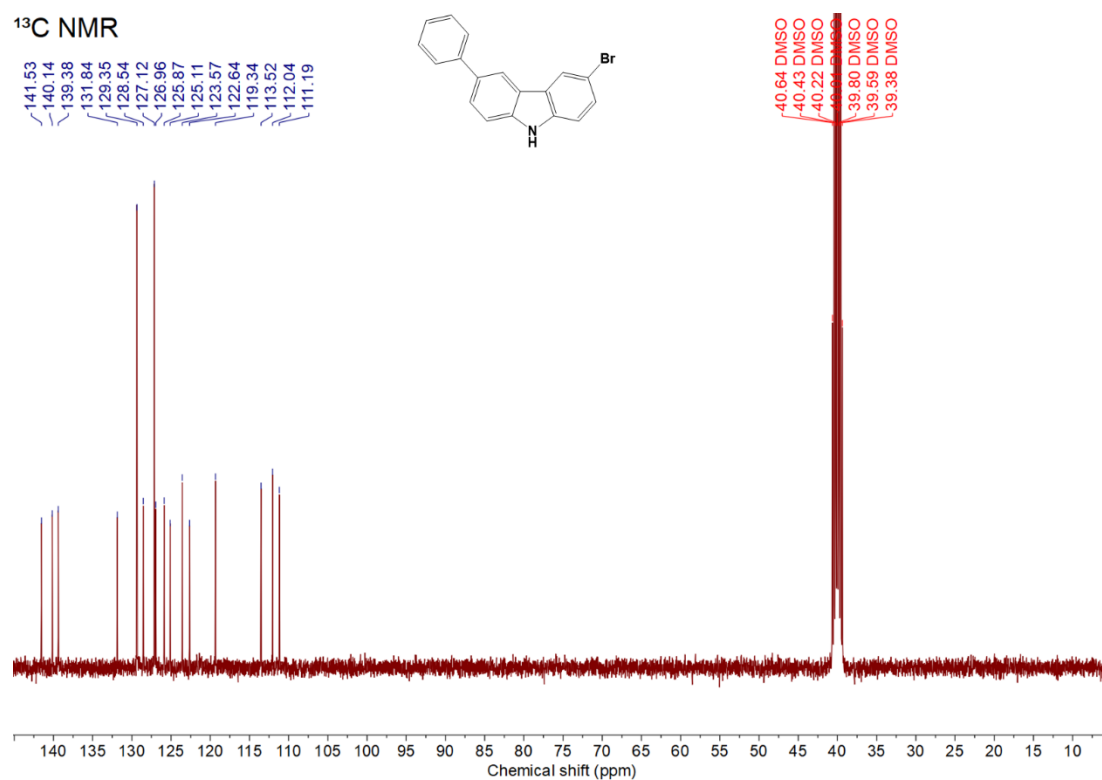
Supplementary Fig. 3. ¹³C NMR of PhBr-4PACz in DMSO.

MS



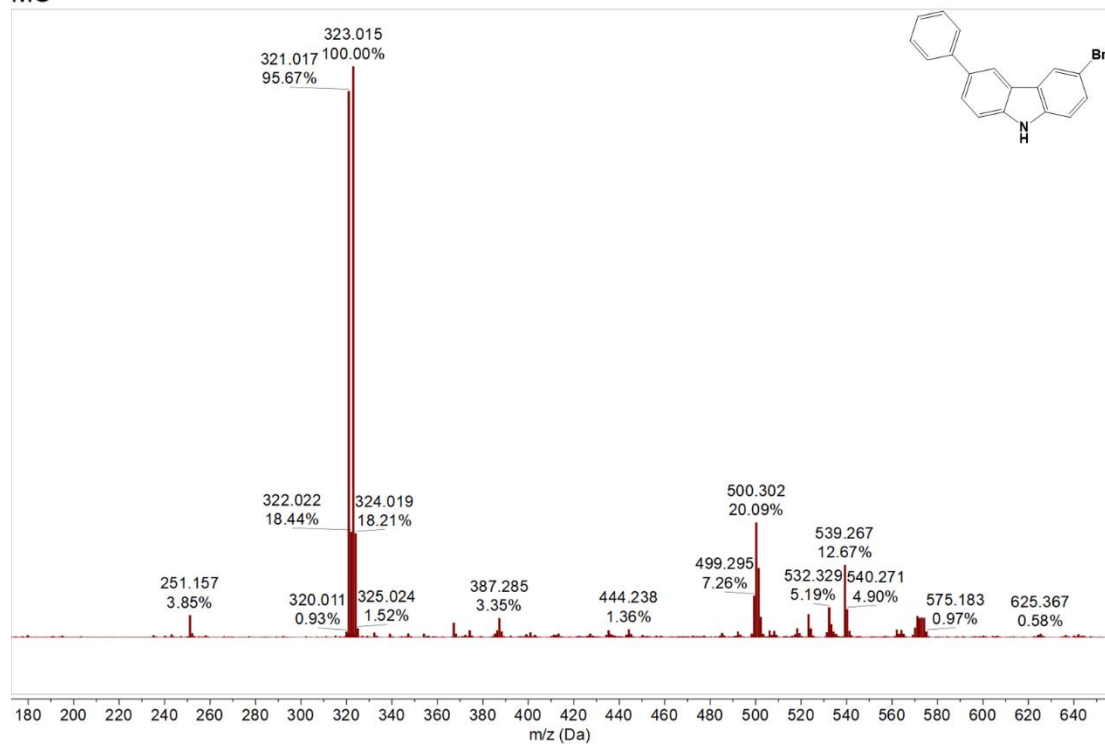


Supplementary Fig. 5. ¹H NMR of 3-bromo-6-phenyl-9H-carbazole in DMSO.



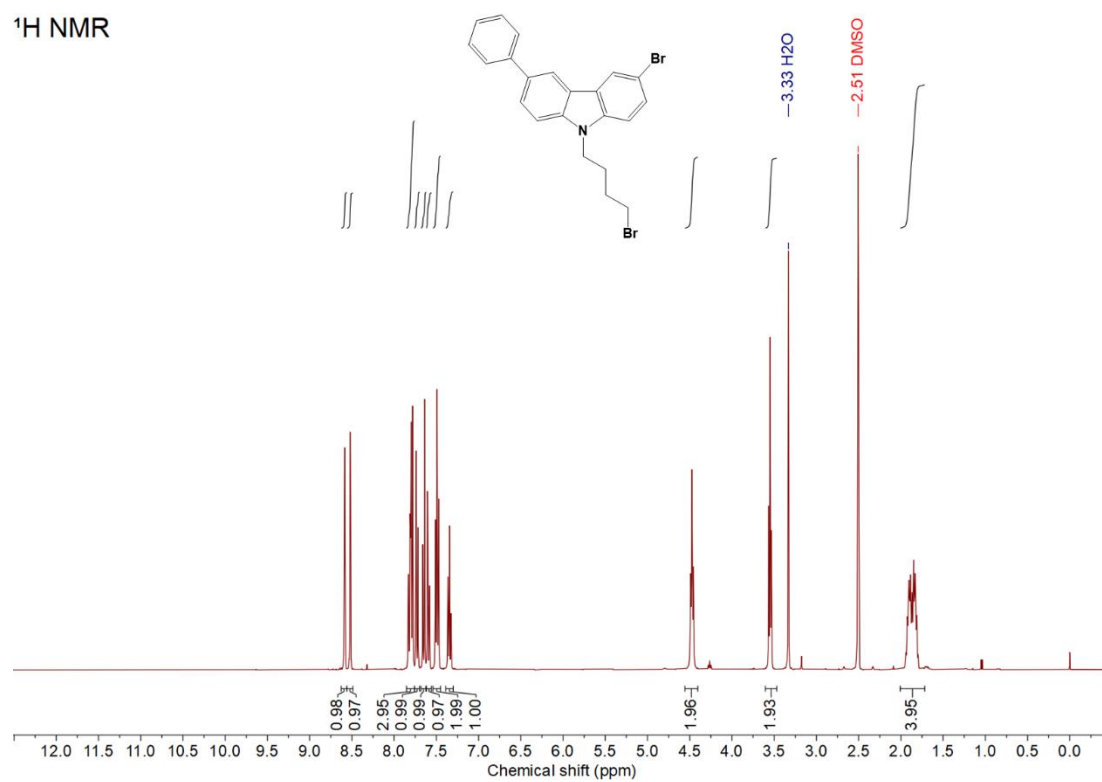
Supplementary Fig. 6. ¹³C NMR of 3-bromo-6-phenyl-9H-carbazole in DMSO.

MS

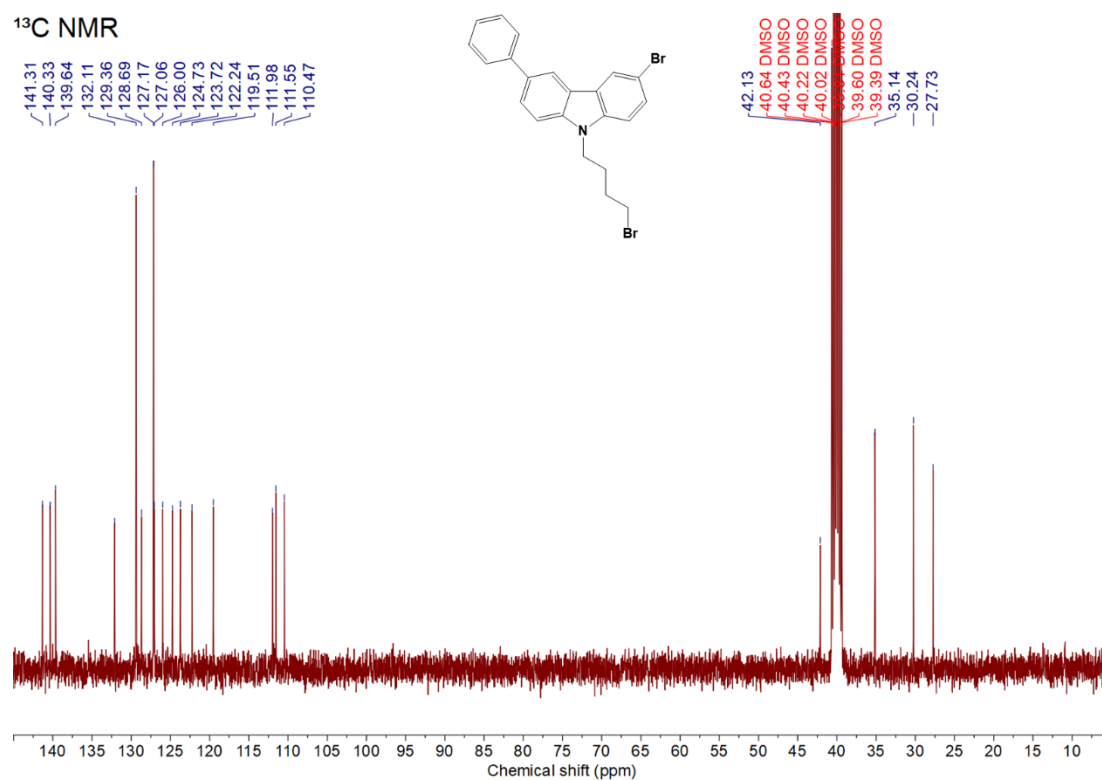


Supplementary Fig. 7. Mass Spectrum of 3-bromo-6-phenyl-9H-carbazole in DMSO.

¹H NMR

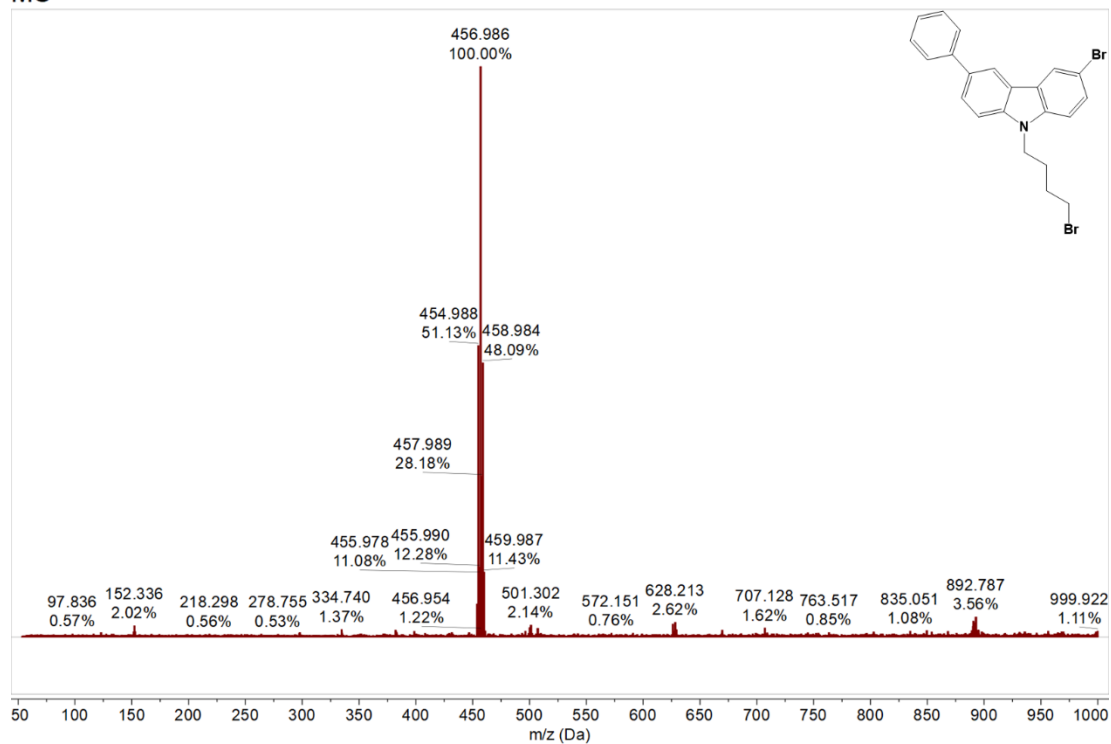


Supplementary Fig. 8. ¹H NMR of 3-bromo-9-(4-bromobutyl)-6-phenyl-9H-carbazole in DMSO.



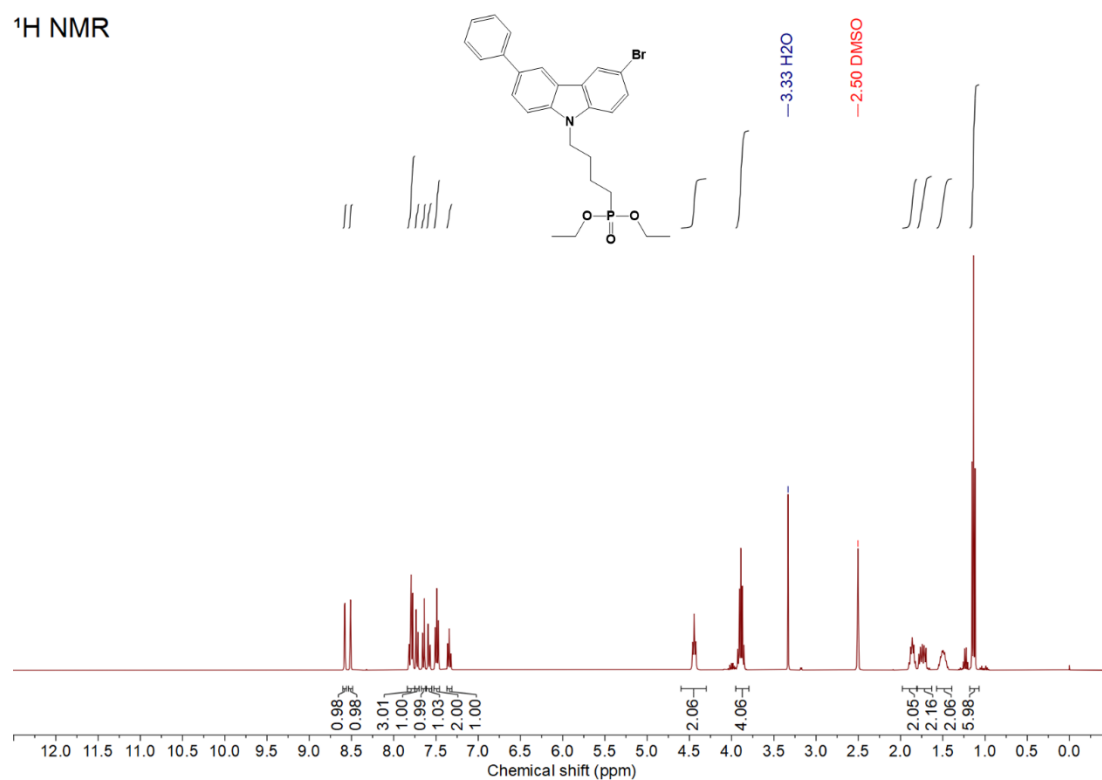
Supplementary Fig. 9. ¹³C NMR of 3-bromo-9-(4-bromobutyl)-6-phenyl-9H-carbazole in DMSO.

MS

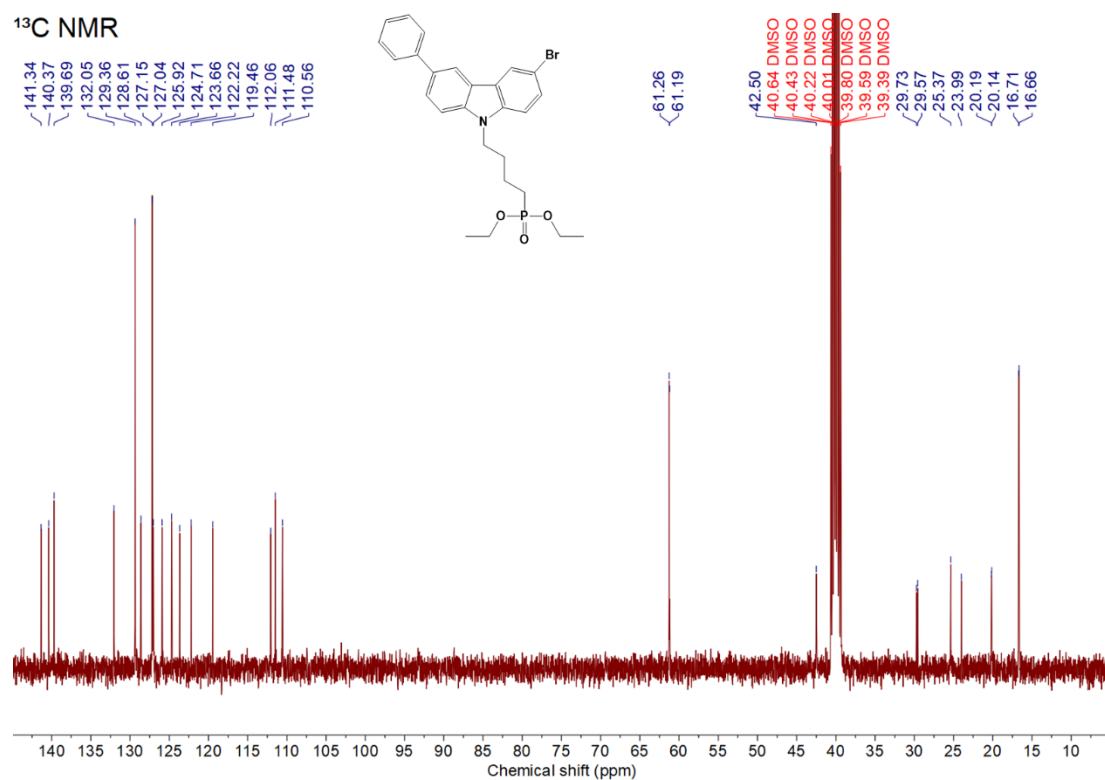


Supplementary Fig. 10. Mass Spectrum of 3-bromo-9-(4-bromobutyl)-6-phenyl-9*H*-carbazole in DMSO.

¹H NMR

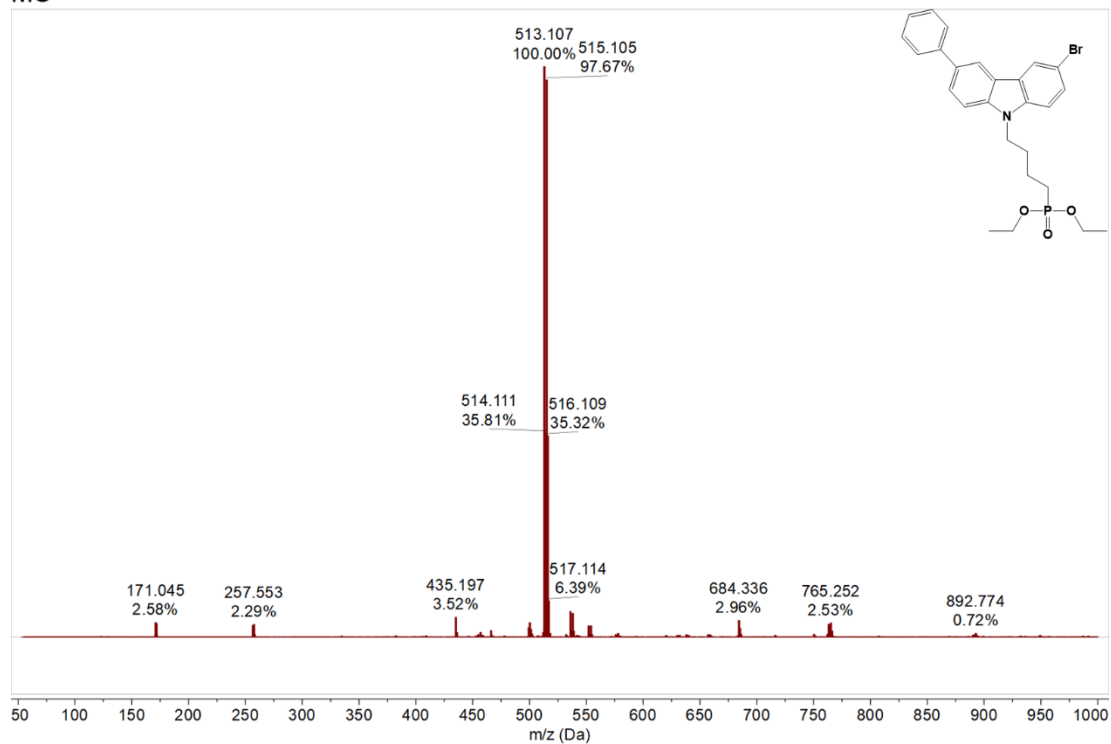


Supplementary Fig. 11. ¹H NMR of diethyl (4-(3-bromo-6-phenyl-9H-carbazol-9-yl)butyl)phosphonate.

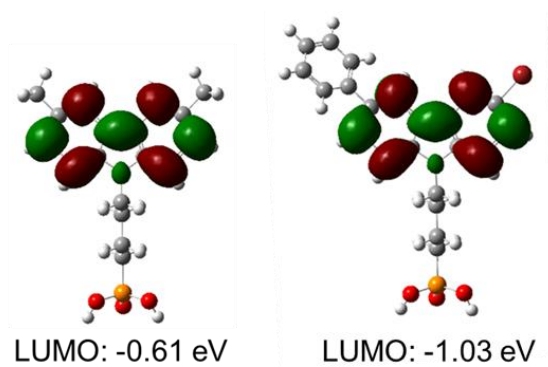


Supplementary Fig. 12. ^{13}C NMR of diethyl (4-(3-bromo-6-phenyl-9H-carbazol-9-yl)butyl)phosphonate in DMSO.

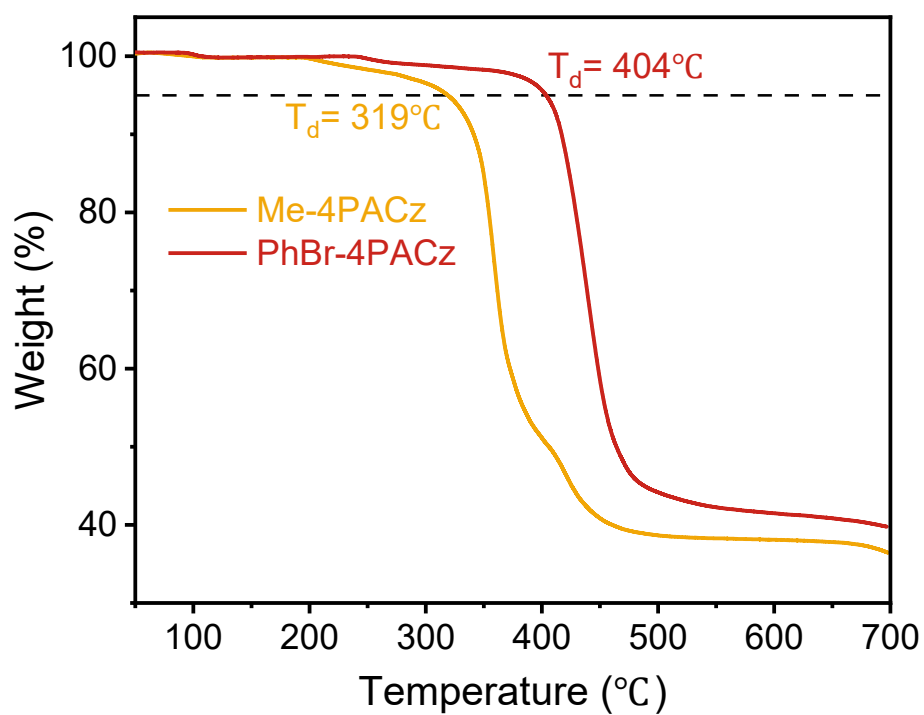
MS



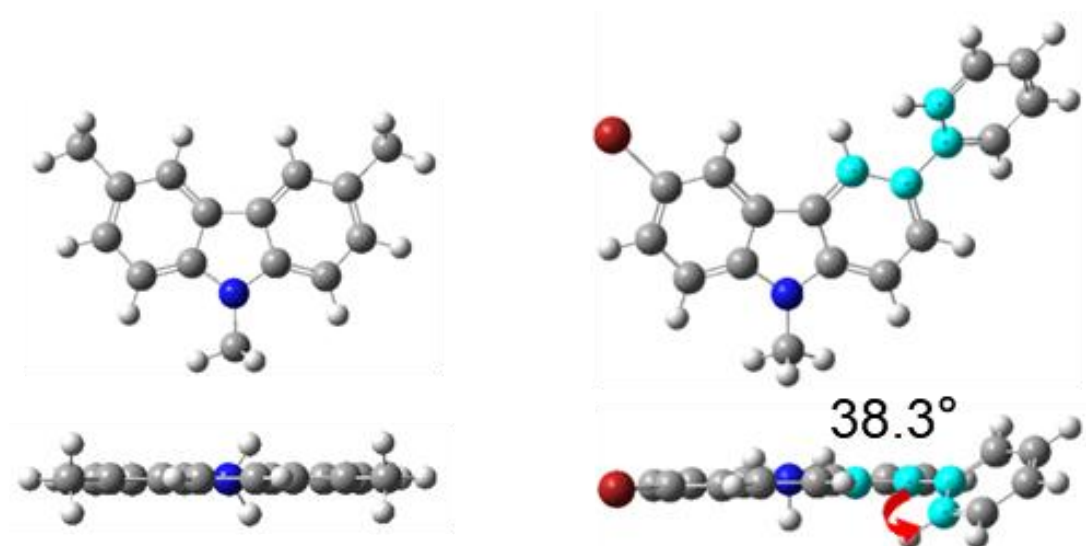
Supplementary Fig. 13. Mass Spectrum of diethyl (4-(3-bromo-6-phenyl-9H-carbazol-9-yl)butyl)phosphonate in DMSO.



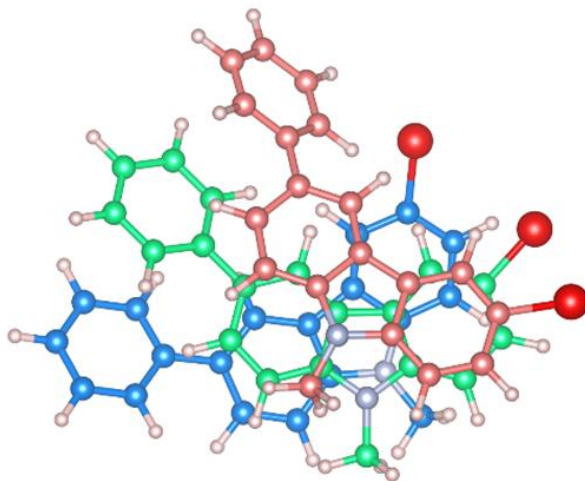
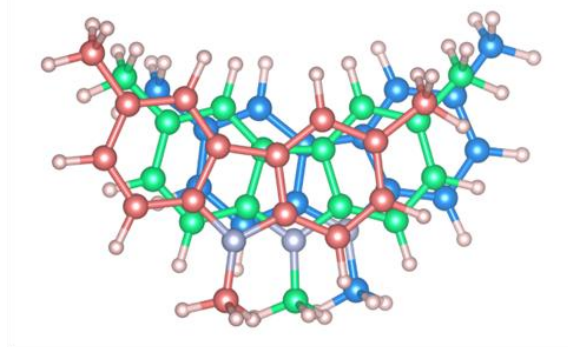
Supplementary Fig. 14. The lowest unoccupied molecular orbital (LUMO) energy levels of Me-4PACz and PhBr-4PACz.



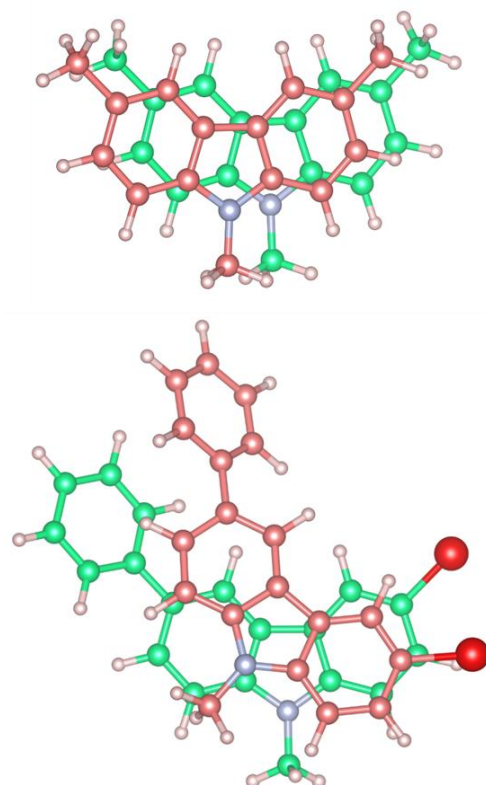
Supplementary Fig. 15. Thermogravimetric analysis curves of Me-4PACz and PhBr-4PACz; heating rate: 20 °C/min, from RT to 700 °C under nitrogen atmosphere.



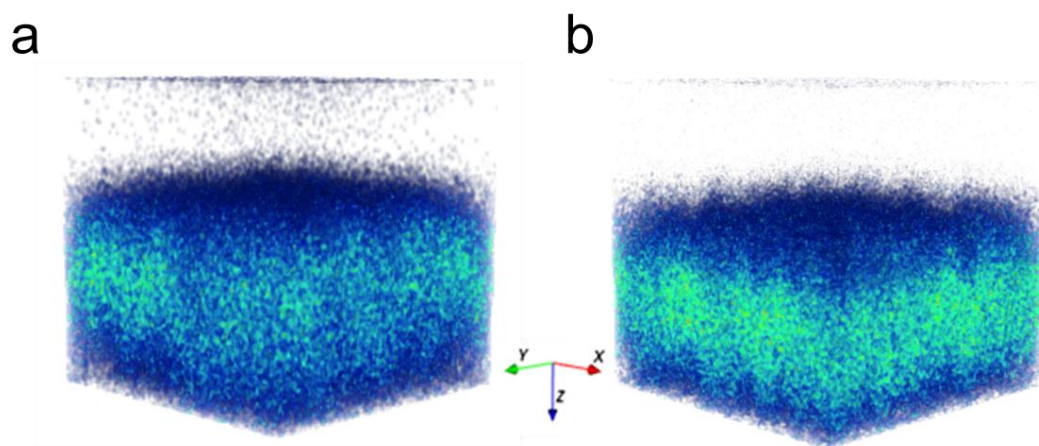
Supplementary Fig. 16. Molecular configuration of the conjugated units for Me-4PACz and PhBr-4PACz.



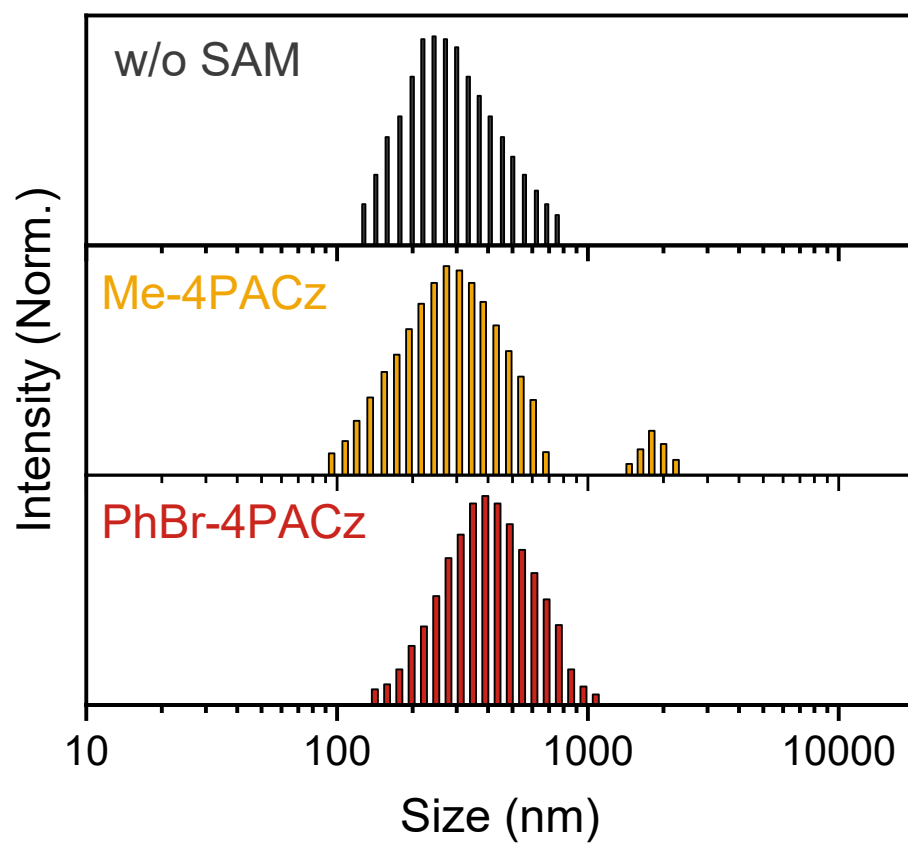
Supplementary Fig. 17. The trimer of the conjugated units for Me-4PACz and PhBr-4PACz.



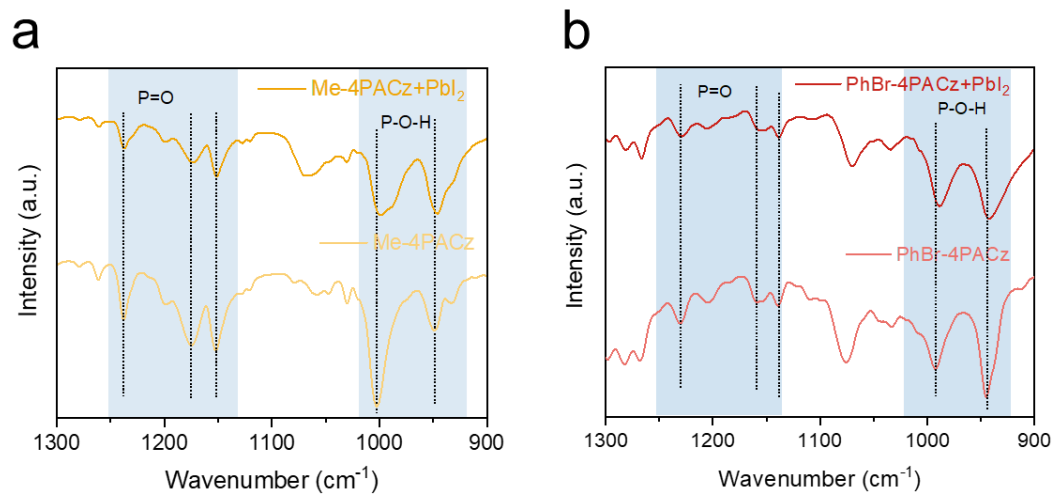
Supplementary Fig. 18. The dimer of the conjugated units of Me-4PACz and PhBr-4PACz.



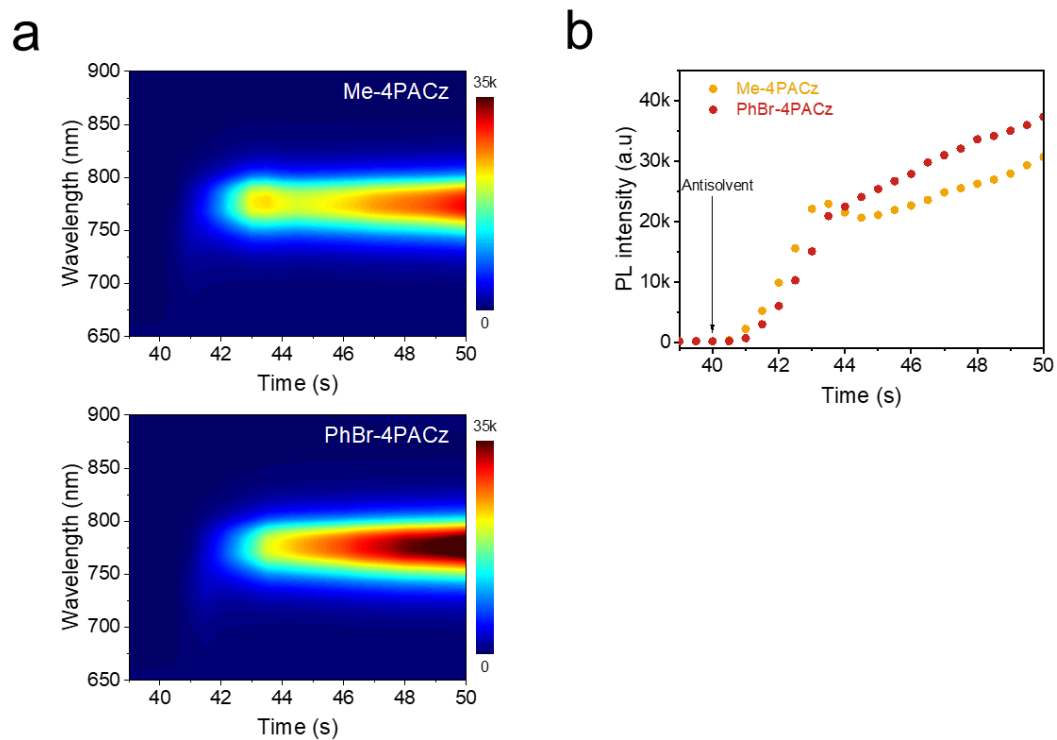
Supplementary Fig. 19. The Spatial distribution of the (a) Me-4PACz (PO_2^-) and (b) PhBr-4PACz (PO_2^-) derived from ToF-SIMS analysis.



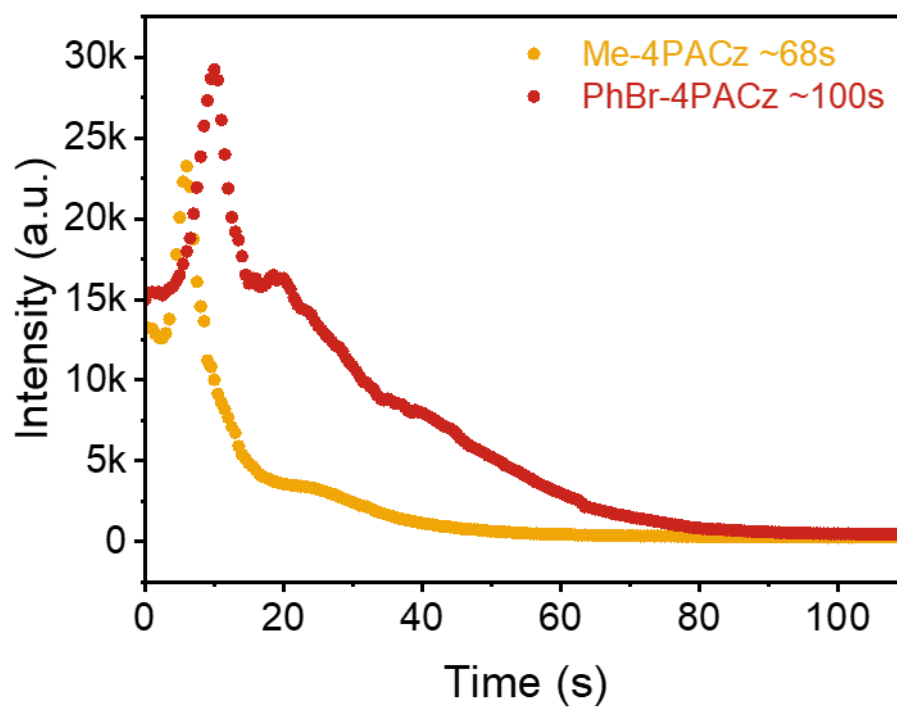
Supplementary Fig. 20. Particle size distribution of precursors with various SAMs as determined by DLS.



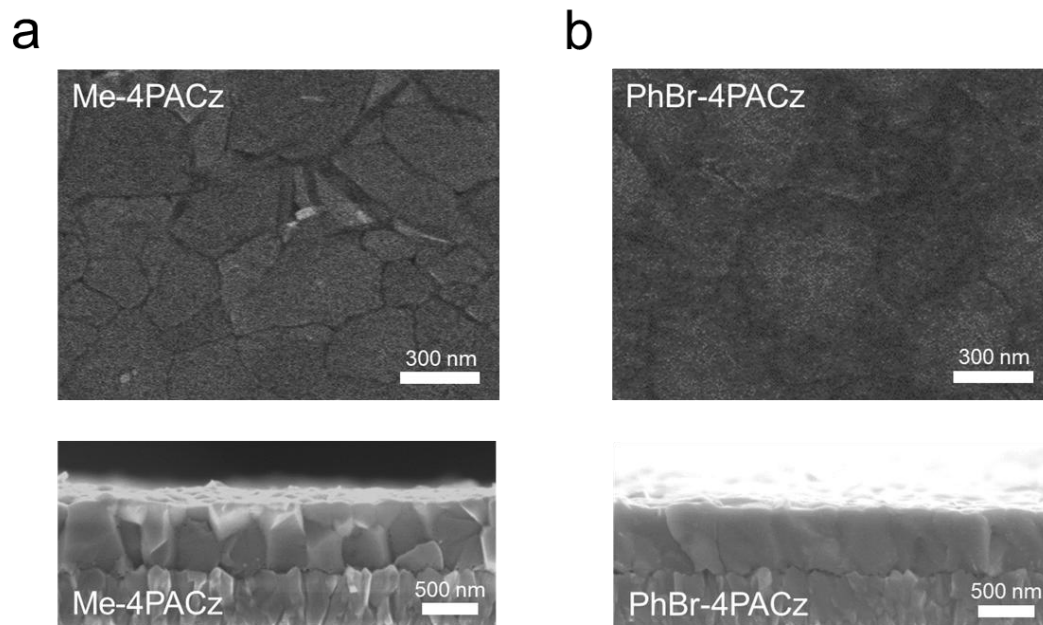
Supplementary Fig. 21. FTIR spectra of Me-4PACz and PhBr-4PACz with PbI₂.



Supplementary Fig. 22. (a) In-situ PL spectra and (b) the corresponding PL peak intensity of Me-4PACz-, and PhBr-4PACz-based samples during the spin-coating process.

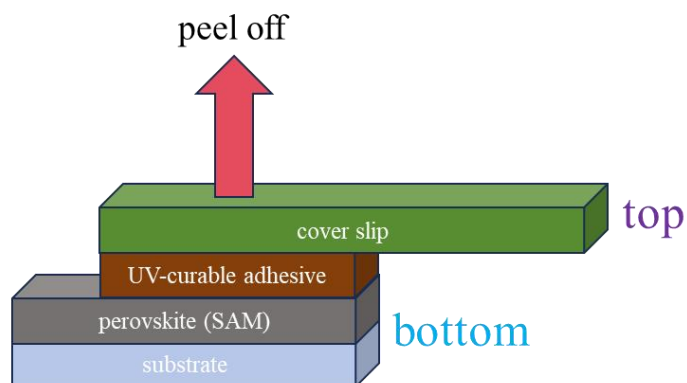


Supplementary Fig. 23. PL peak intensity of Me-4PACz-, and PhBr-4PACz-based samples during the annealing process.

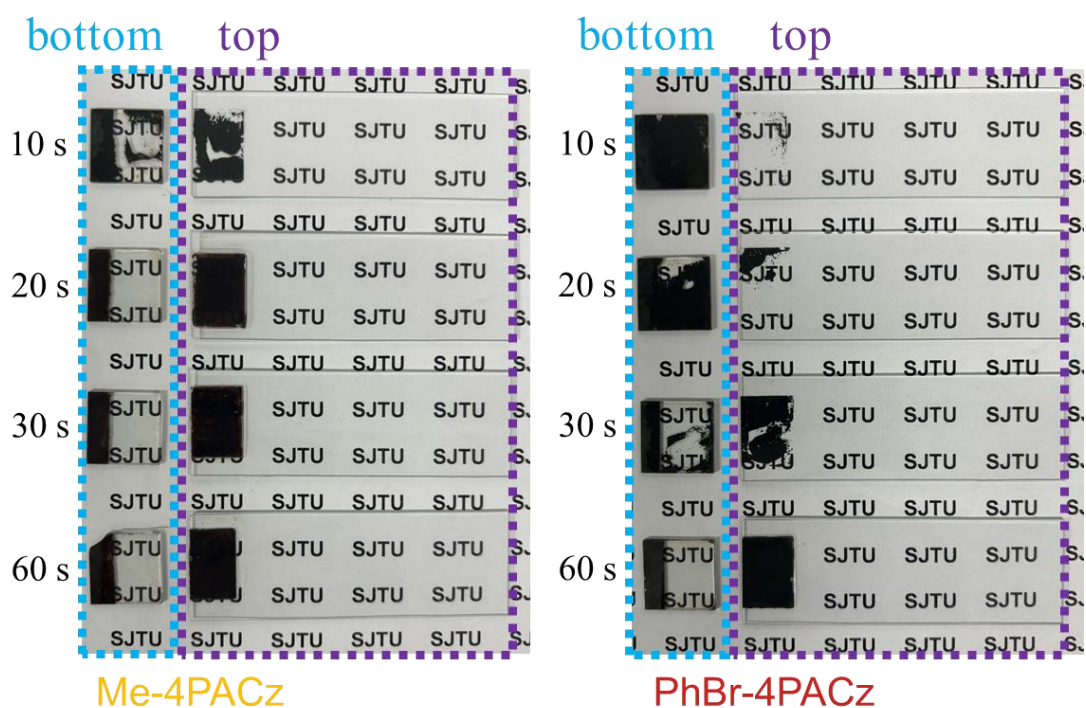


Supplementary Fig. 24. SEM images of (a) perovskite (Me-4PACz) and (b) perovskite (PhBr-4PACz) films, including buried surface and cross-section views.

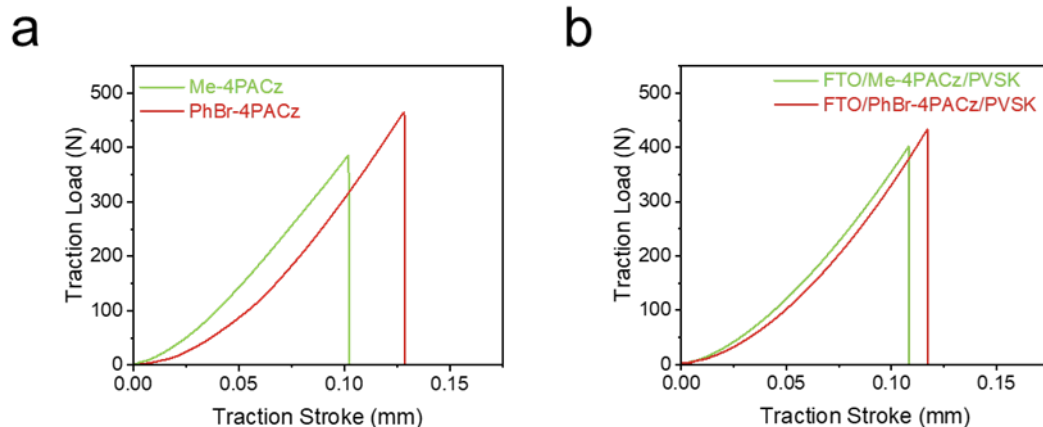
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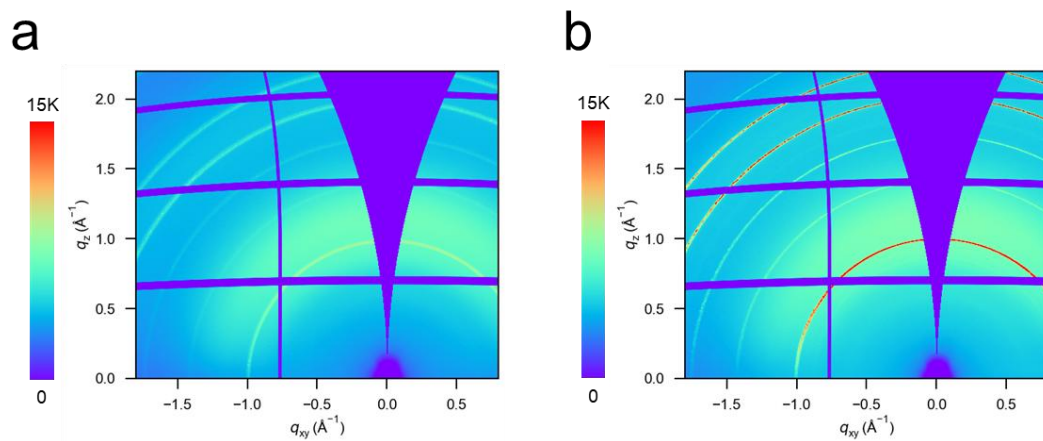
b



Supplementary Fig. 25. (a) Schematic diagram of the interface stripping device structure. Perovskite (SAM) films were deposited on substrates. UV-curable adhesive was then deposited on the surface of the perovskite (SAM) films, followed by bonding with a cover slip using ultraviolet (UV) irradiation for a certain period of time. (b) After irradiation and curing, the cover slip was peeled off, separating the poorly bonded sections of the perovskite (SAM) film from the substrate.

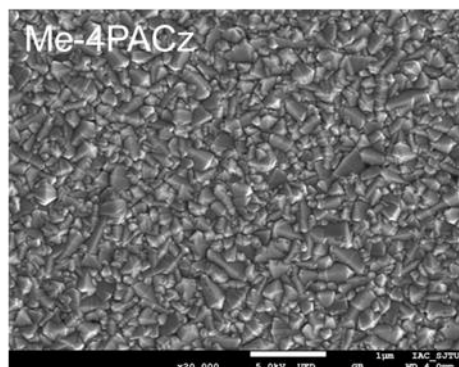


Supplementary Fig. 26. Stress-strain curves of the (a) perovskite (SAM) film prepared by the co-deposited process and (b) SAM/perovskite film prepared by the sequential-deposited process.

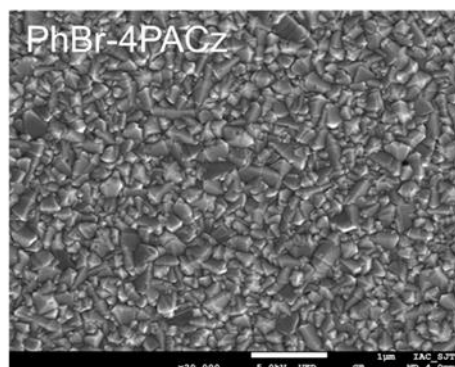


Supplementary Fig. 27. GIWAXS images of the buried surface of (a) perovskite (Me-4PACz) and (b) perovskite (PhBr-4PACz) films.

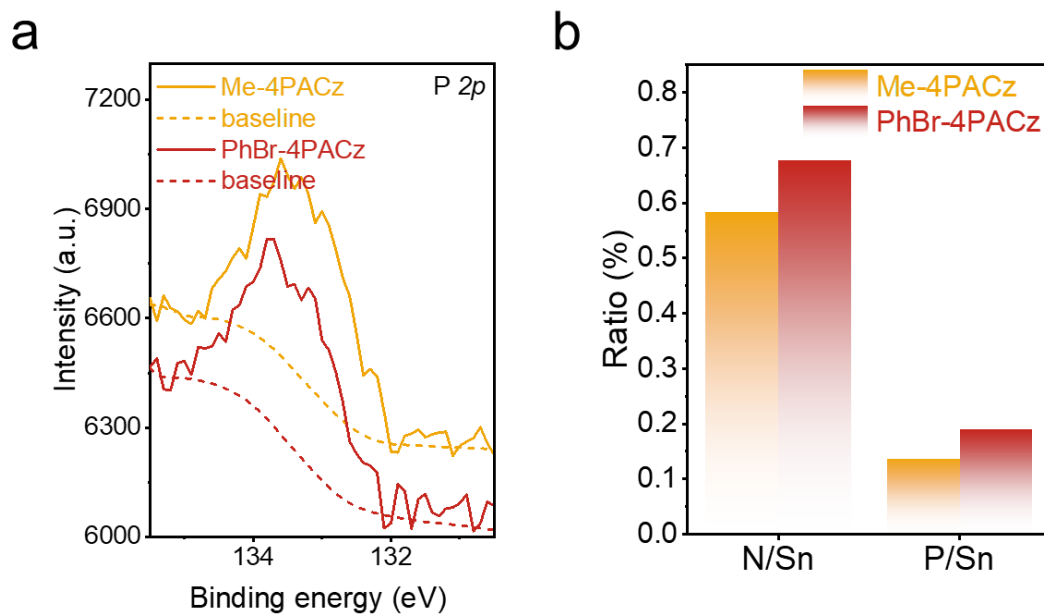
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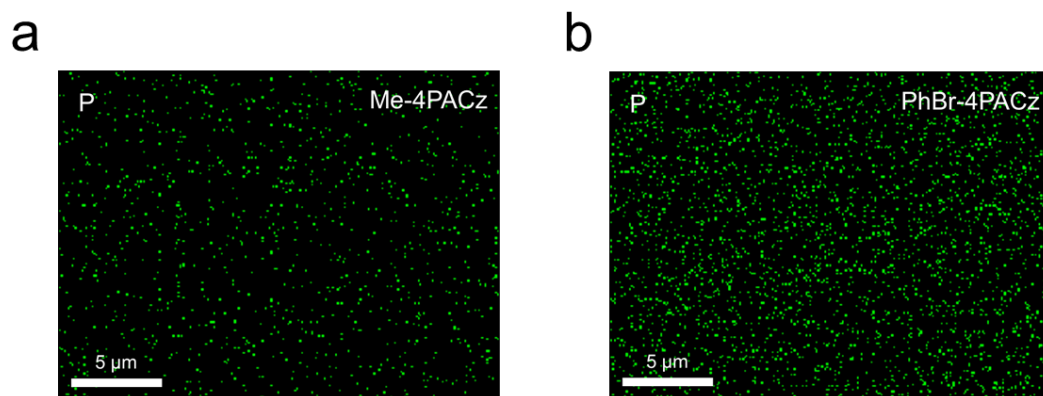
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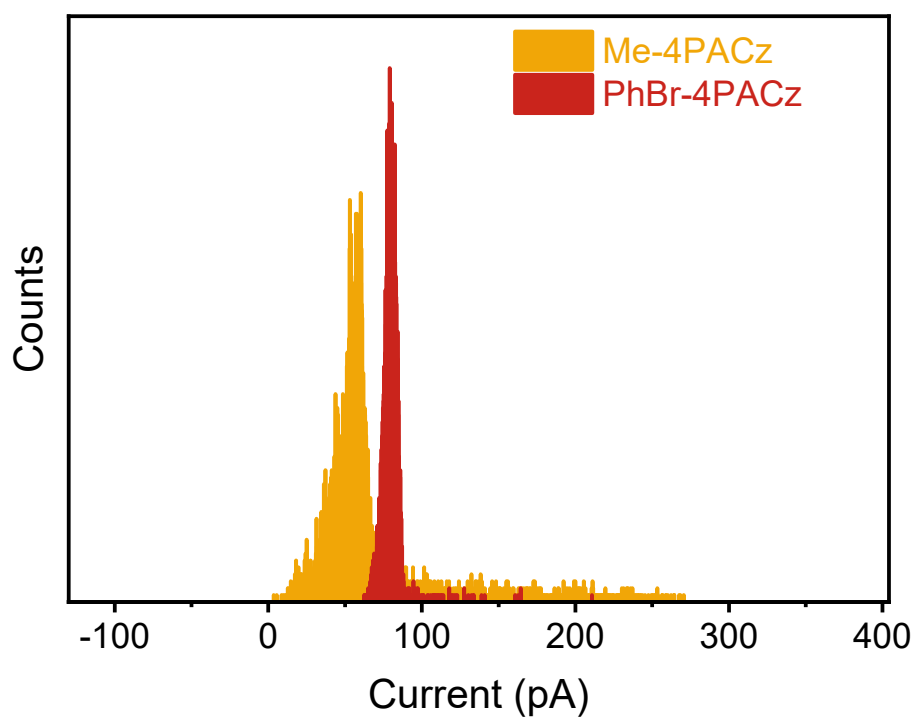
Supplementary Fig. 28. Top view SEM images for the FTO/SAM substrates after washing off the (a) perovskite (Me-4PACz) film and (b) perovskite (Me-4PACz) film.



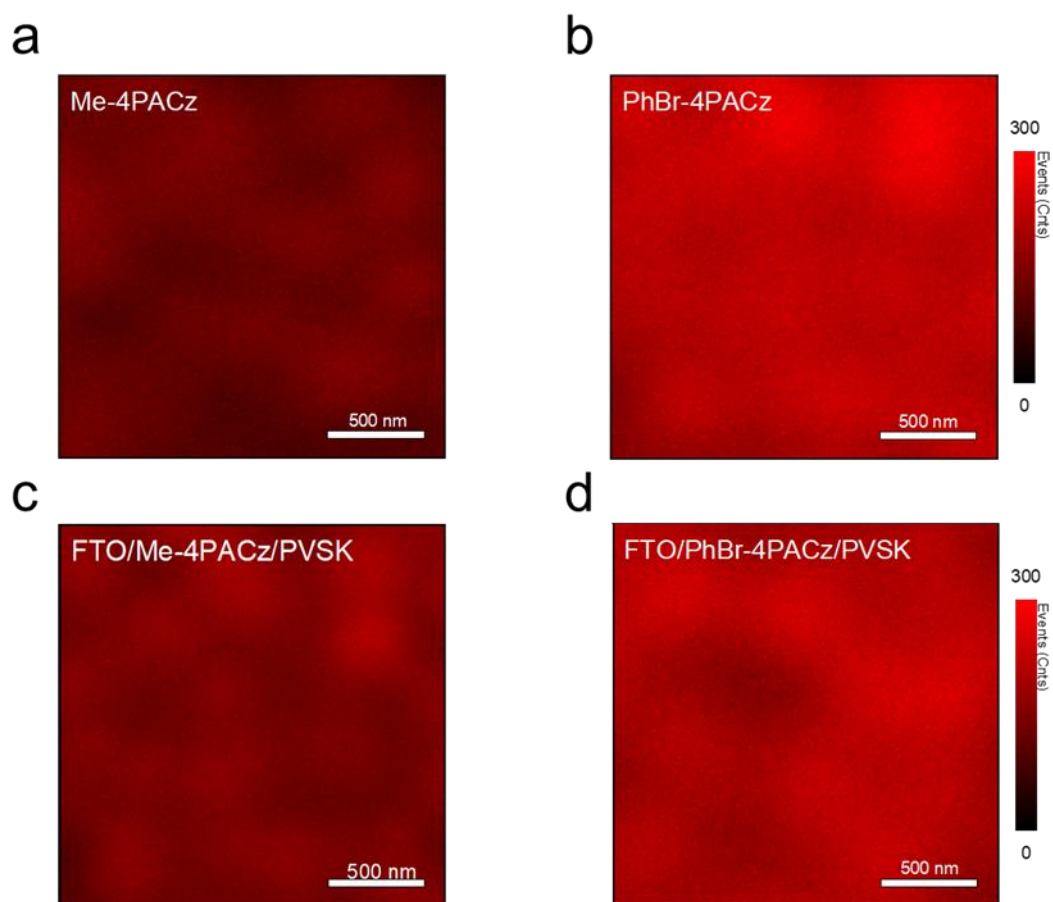
Supplementary Fig. 29. (a) P 2p XPS spectra for the FTO/SAM substrates after washing off the perovskite (SAM) films. (b) The N/Sn and P/Sn area ratios calculated based on XPS data.



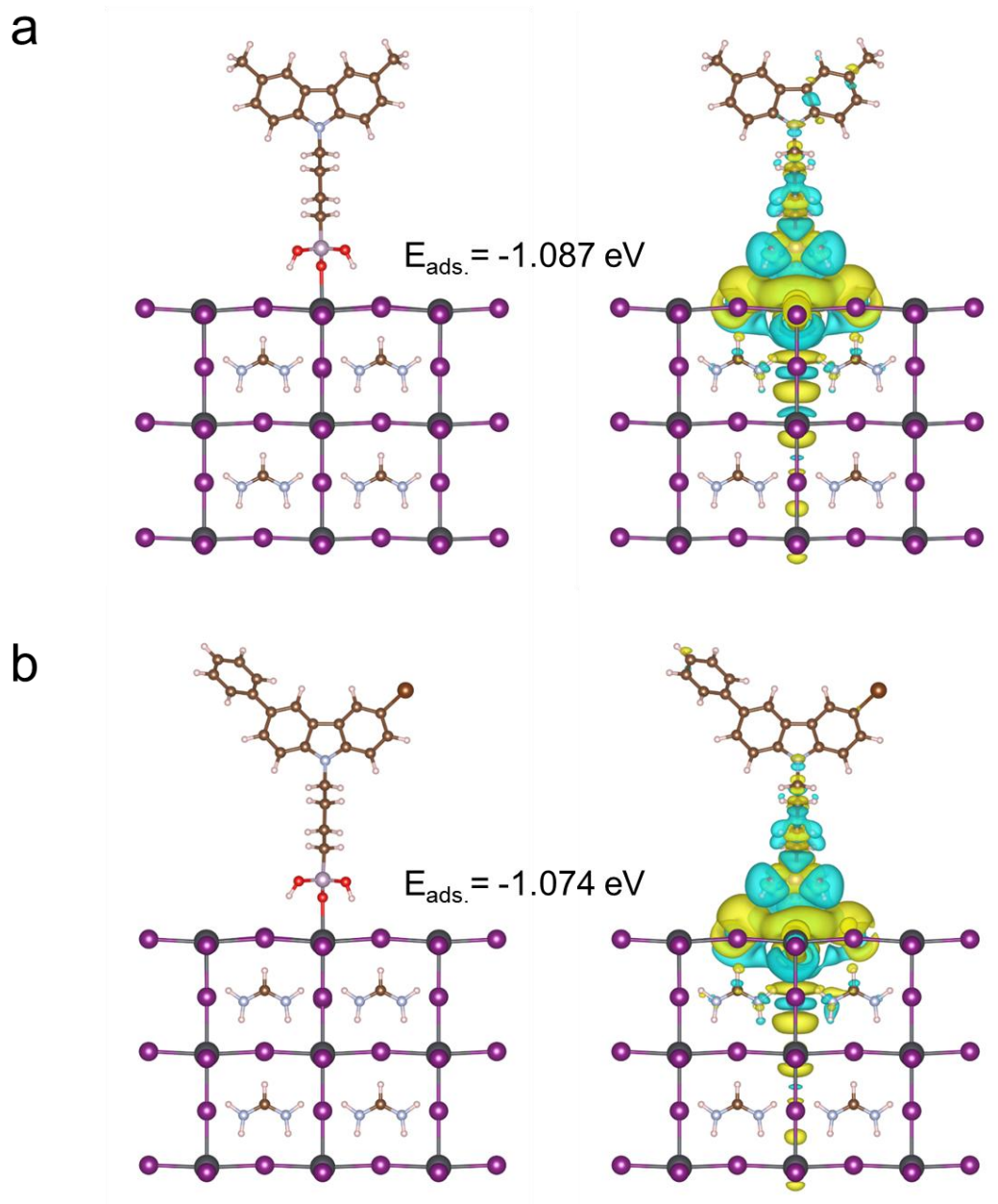
Supplementary Fig. 30. Elemental mapping of P for the (a) FTO/Me-4PACz and (b) FTO/PhBr-4PACz substrates after washing off the perovskite (SAM) films.



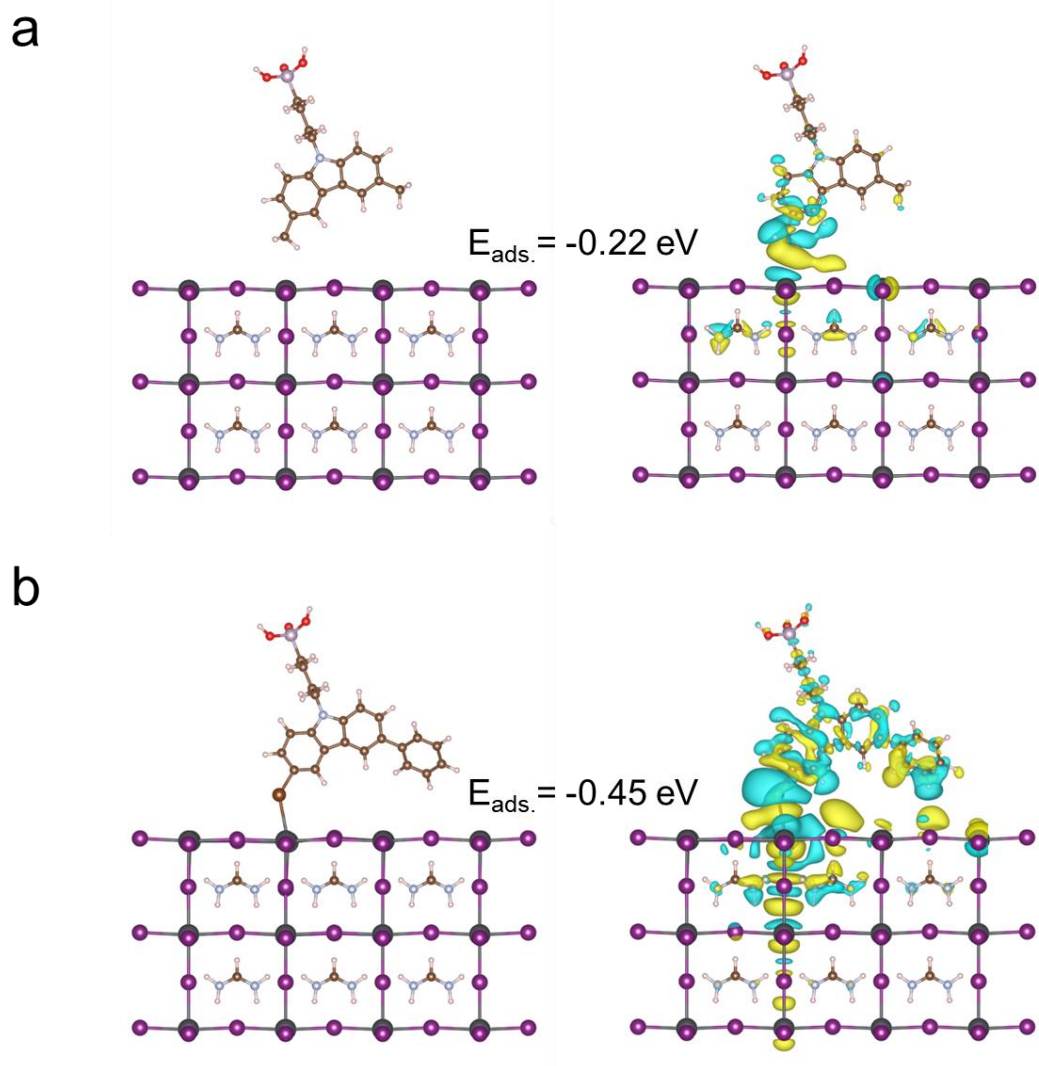
Supplementary Fig. 31. The statistical distribution of the current signal for the FTO/Me-4PACz and FTO/PhBr-4PACz substrates after washing off the perovskite (SAM) films.



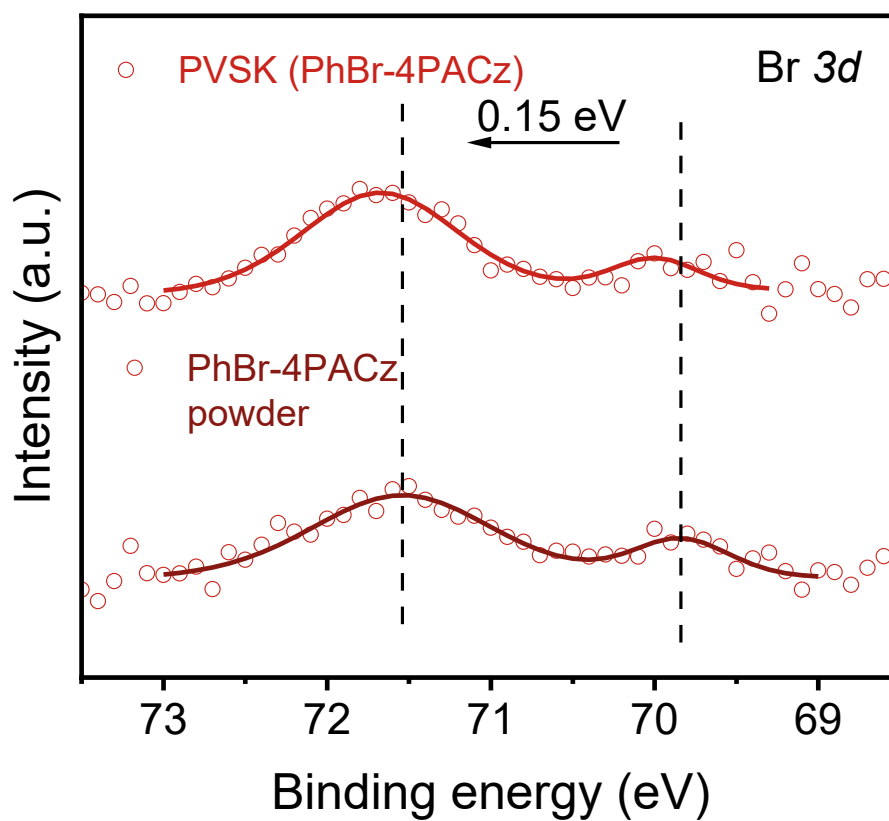
Supplementary Fig. 32. Photoluminescence (PL) mapping images of perovskite films: (a) with Me-4PACz and (b) with PhBr-4PACz prepared by the co-deposited process; (c) Me-4PACz/perovskite and (d) PhBr-4PACz/perovskite prepared by the sequential-deposited process. The images were collected from the bottom side.



Supplementary Fig. 33. The optimized structure for the adsorption of (a) Me-4PACz and (b) PhBr-4PACz on the perovskite surface by anchoring groups and the corresponding charge density difference distribution.



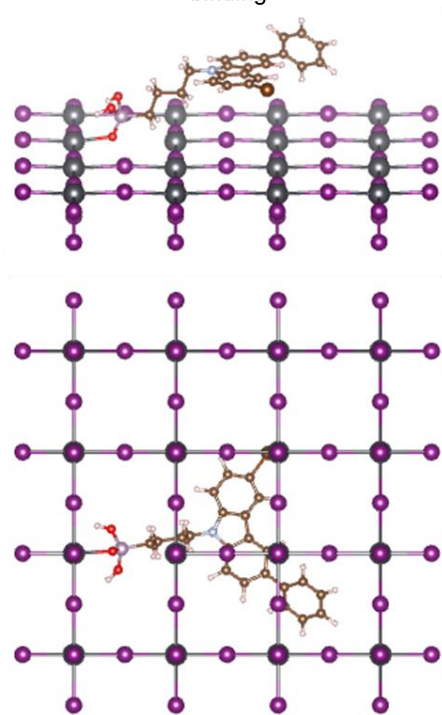
Supplementary Fig. 34. The optimized structure for the adsorption of (a) Me-4PACz and (b) PhBr-4PACz on the perovskite surface by terminal groups and the corresponding charge density difference distribution.



Supplementary Fig. 35. XPS spectra of Br 3d for buried surface of perovskite (PhBr-4PACz) film and pure PhBr-4PACz powder.

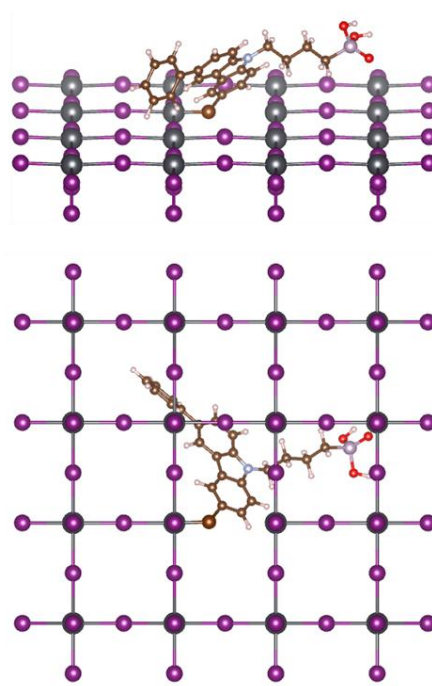
a

$$E_{\text{binding}} = -1.28 \text{ eV}$$



b

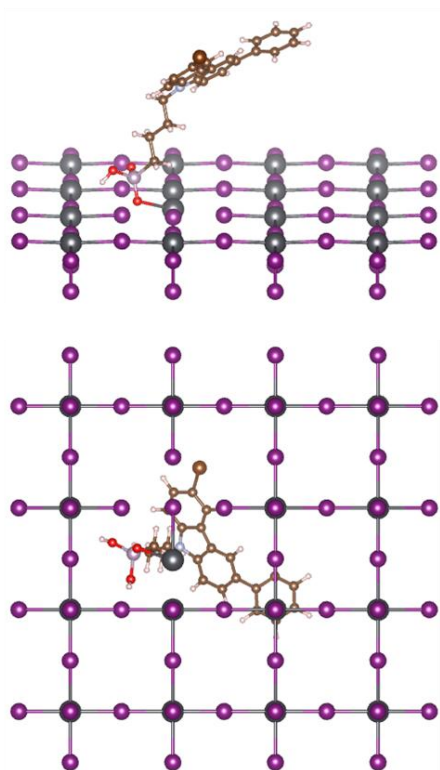
$$E_{\text{binding}} = -1.20 \text{ eV}$$



Supplementary Fig. 36. Theoretical models of perovskite with molecular surface passivation of V_I defects with (a) PA group and (b) Br group of PhBr-4PACz.

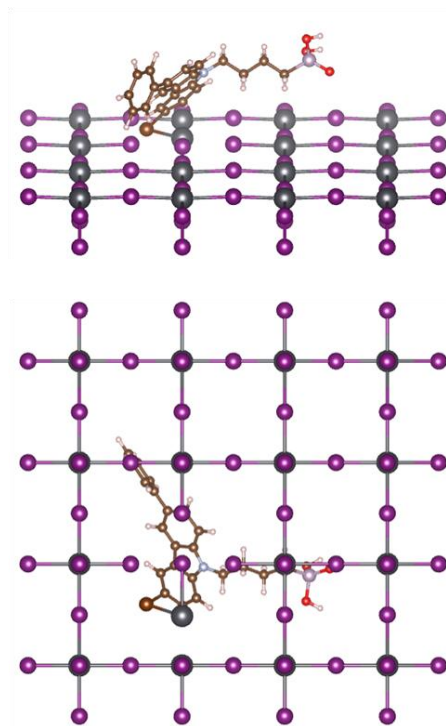
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$$E_{\text{binding}} = -1.55 \text{ eV}$$

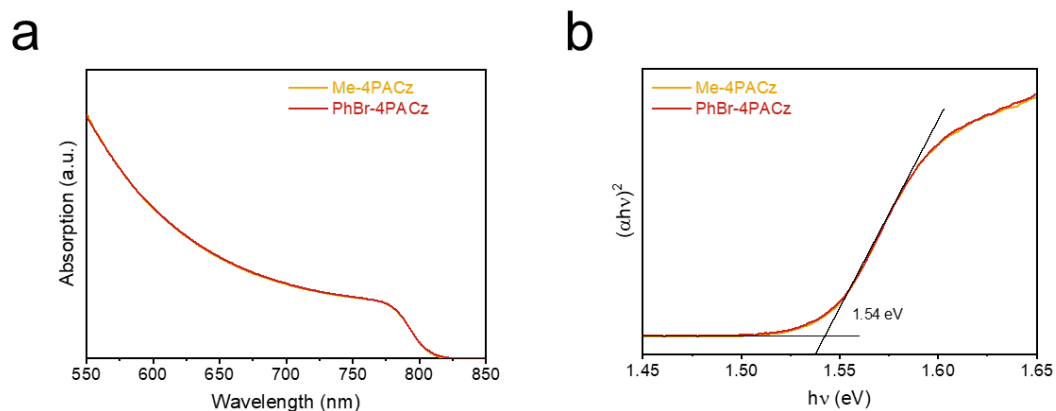


b

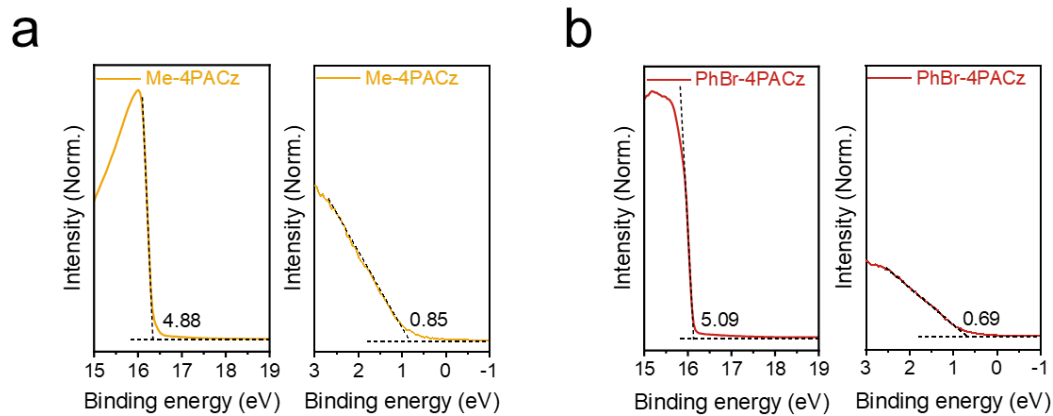
$$E_{\text{binding}} = -1.29 \text{ eV}$$



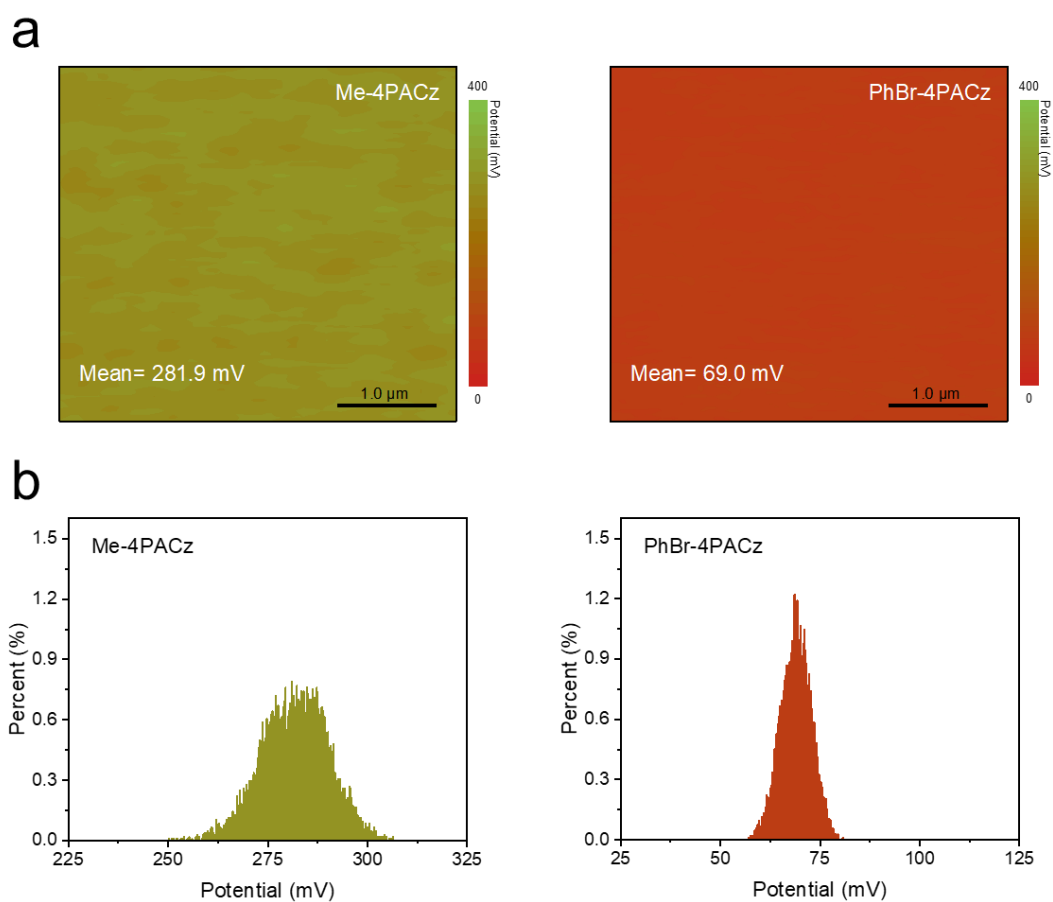
Supplementary Fig. 37. Theoretical models of perovskite with molecular surface passivation of Pb_I defects with (a) PA group and (b) Br group of PhBr-4PACz.



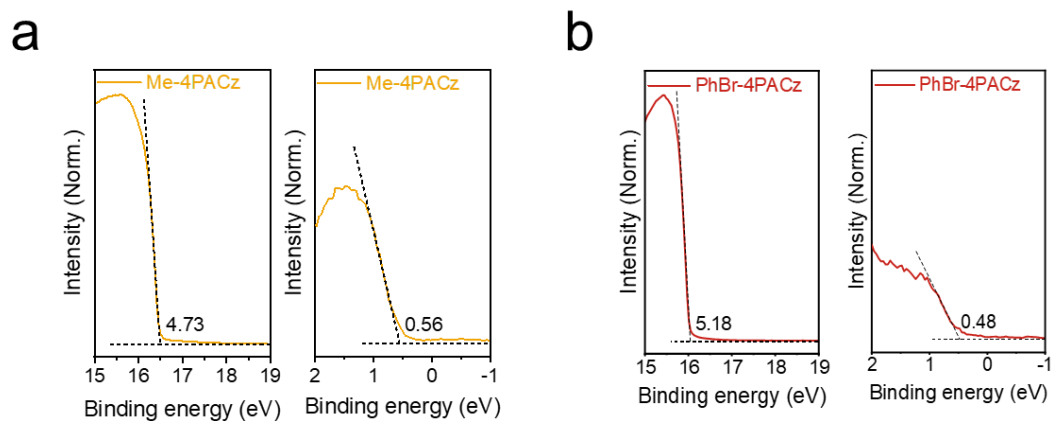
Supplementary Fig. 38. (a) UV-vis absorption spectrum of perovskite (Me-4PACz) and perovskite (PhBr-4PACz) films and (b) optical bandgap calculated from a Tauc plot with absorption coefficient versus photon energy.



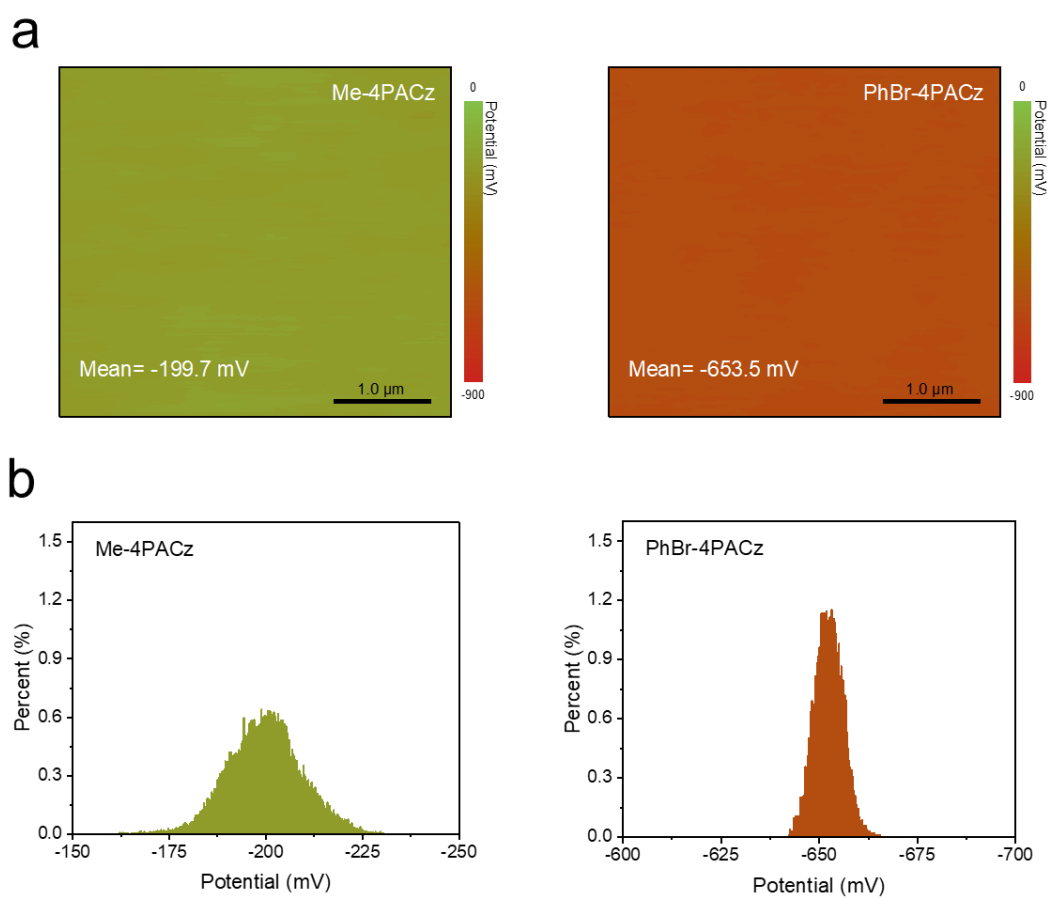
Supplementary Fig. 39. UPS spectra of secondary electron cutoff region and Fermi edge region for bottom surface of (a) perovskite (Me-4PACz) film and (b) perovskite (PhBr-4PACz) film.



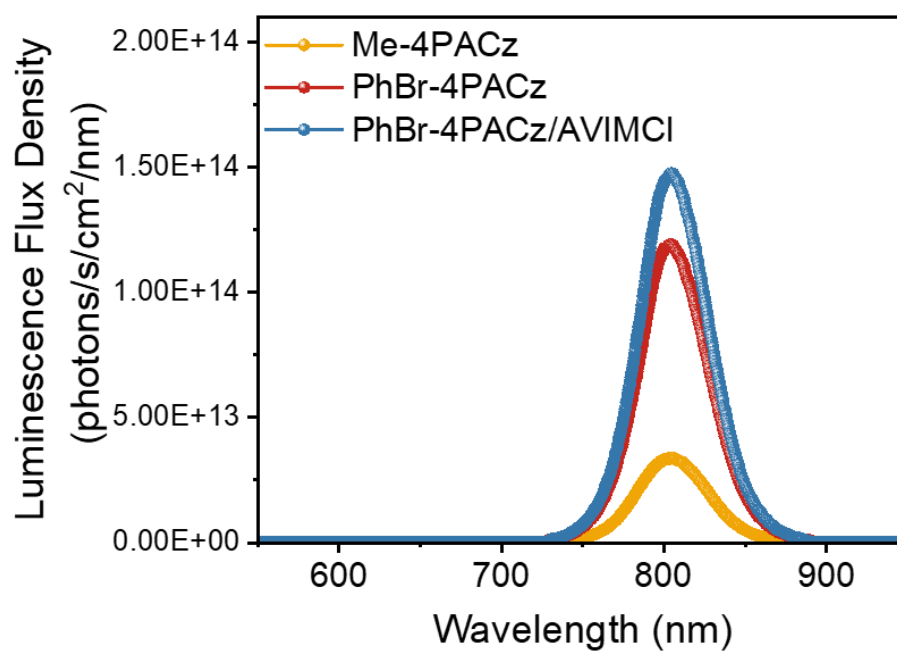
Supplementary Fig. 40. (a) KPFM images and (b) corresponding statistics of surface potential distribution of bottom surface of perovskite (Me-4PACz) film and perovskite (PhBr-4PACz) film.



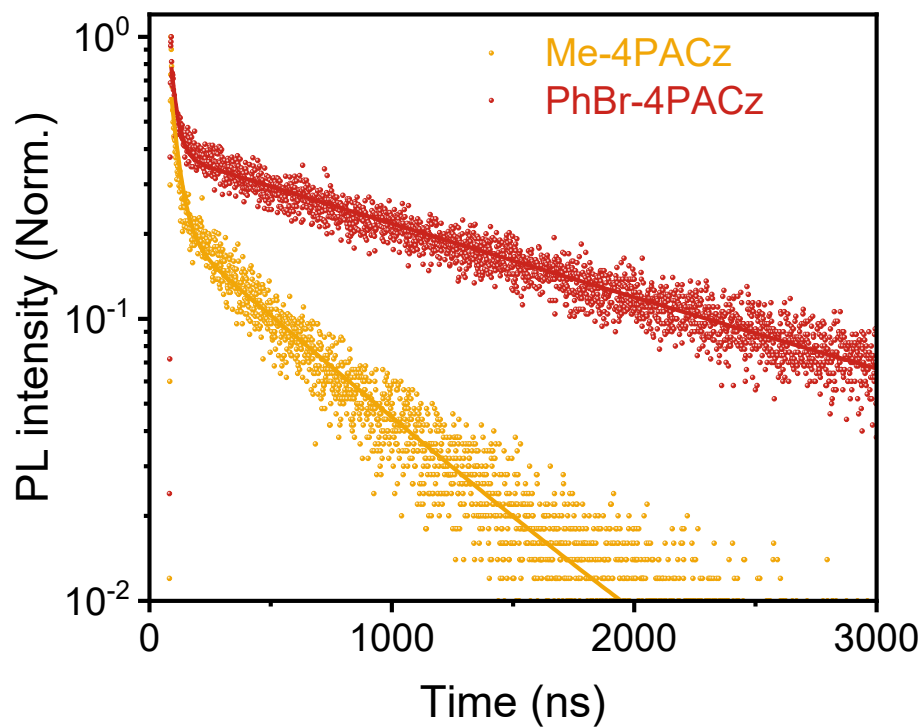
Supplementary Fig. 41. UPS spectra of secondary electron cutoff region and Fermi edge region for (a) FTO/Me-4PACz substrate (b) FTO/PhBr-4PACz substrate after washing off the perovskite (SAM) film.



Supplementary Fig. 42. (a) KPFM images and (b) corresponding statistics of surface potential distribution of FTO/Me-4PACz substrate and FTO/PhBr-4PACz substrate after washing off the perovskite (SAM) film.

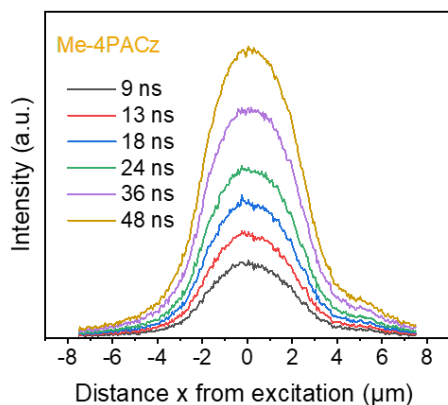


Supplementary Fig. 43. The absolute photoluminescence spectrum of perovskite films with Me-4PACz, PhBr-4PACz and PhBr-4PACz/AVIMCl.

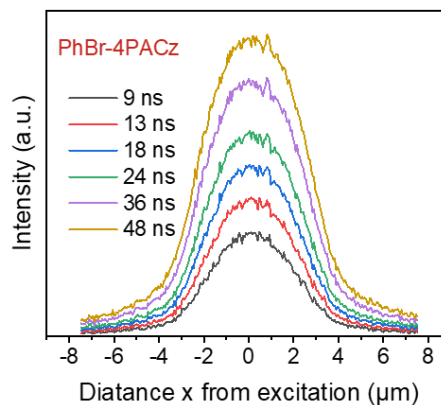


Supplementary Fig. 44. TRPL decays of perovskite (Me-4PACz) film and perovskite (PhBr-4PACz) film. The data were collected from the bottom side.

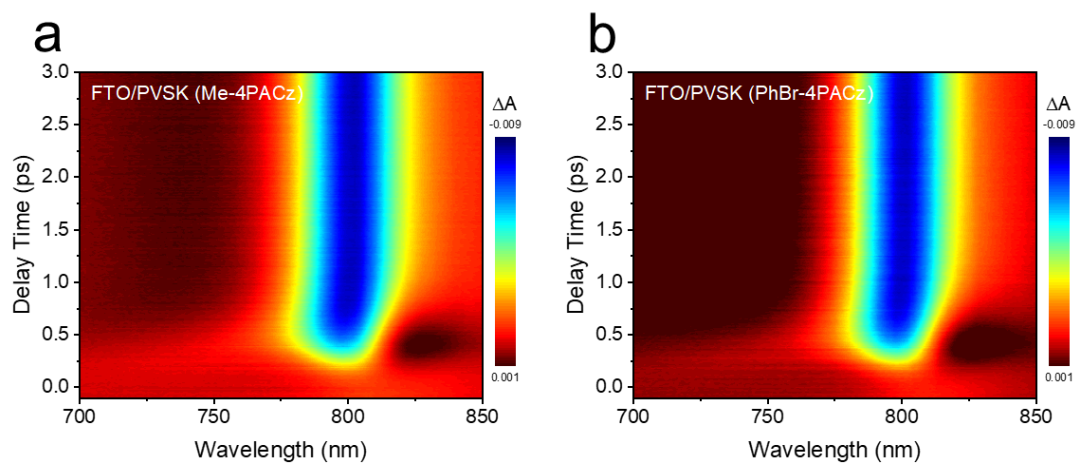
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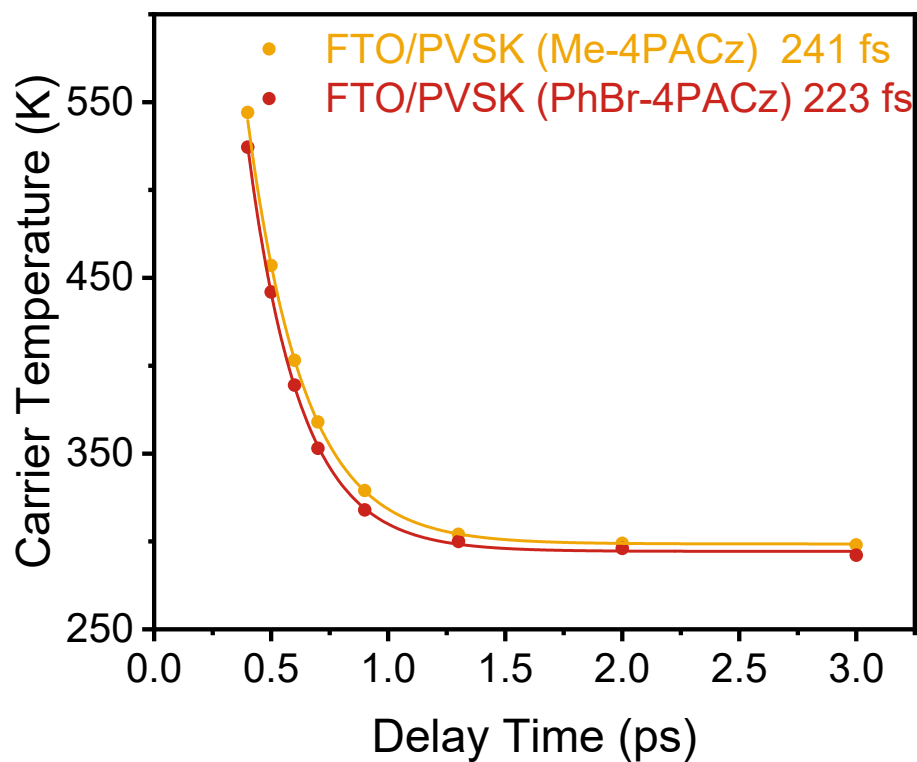
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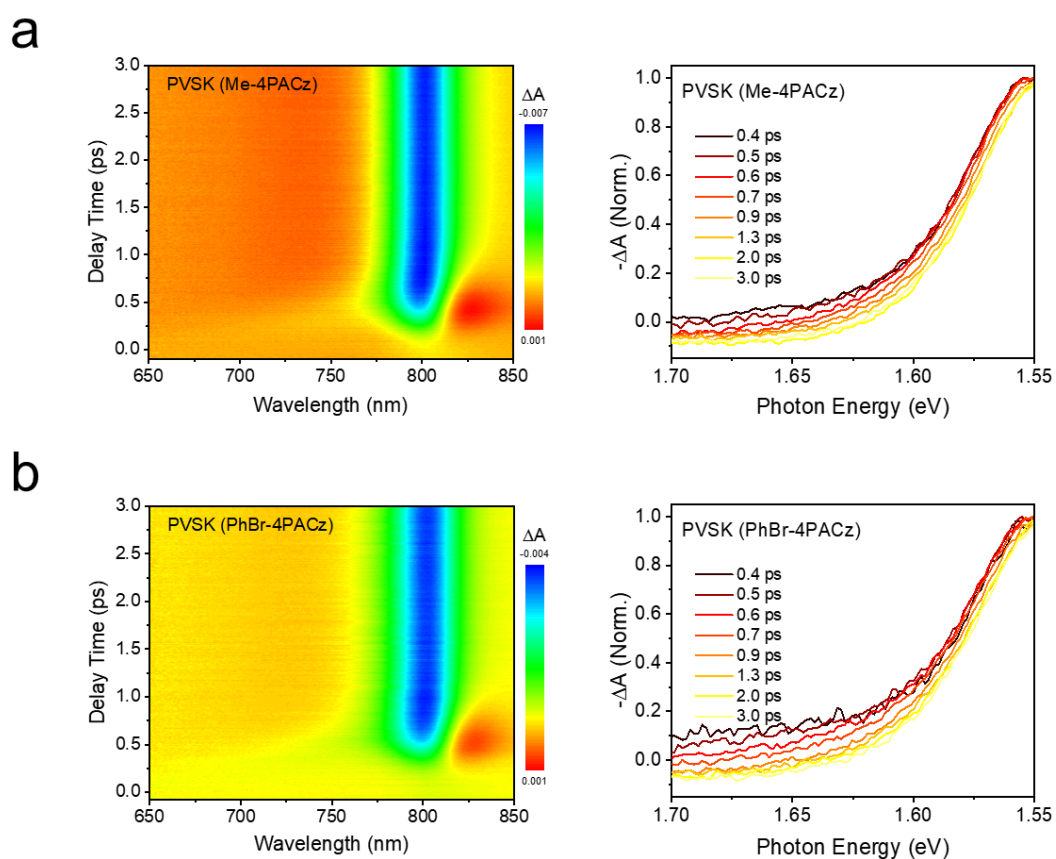
Supplementary Fig. 45. The PL curves corresponding to the (a) perovskite (Me-4PACz) film and (b) perovskite (PhBr-4PACz) film at different snapshot times t . By performing Gaussian fitting on the PL curves, the standard deviation $\sigma_x(t)$ is obtained, which was used to evaluate carrier diffusion (D).



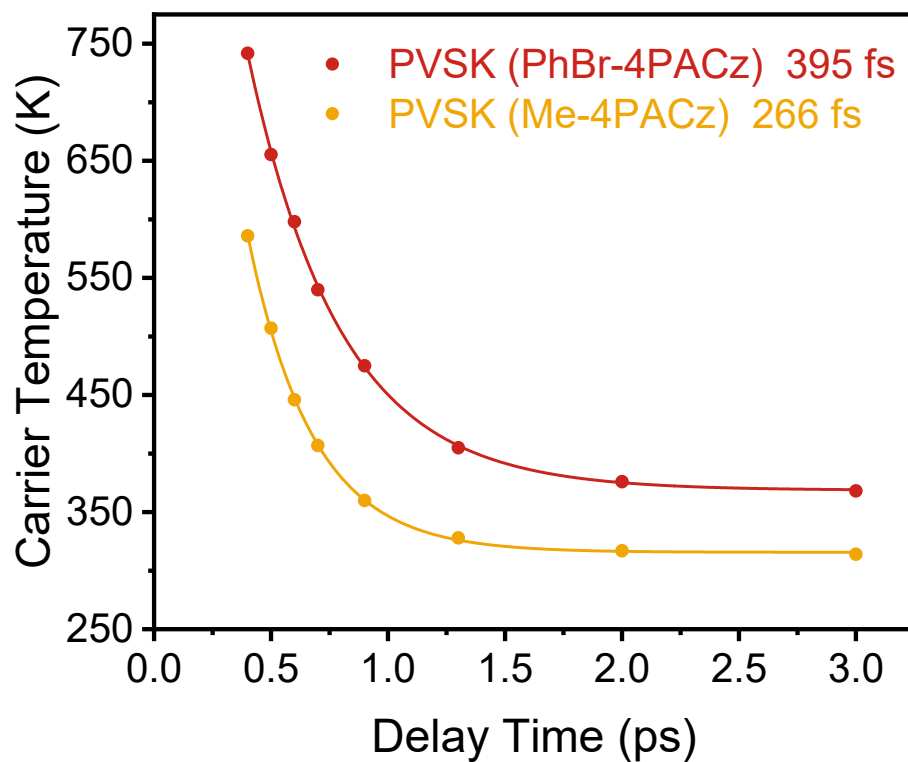
Supplementary Fig. 46. Pseudo color plot of (a) perovskite (Me-4PACz) film and (b) perovskite (PhBr-4PACz) film deposited on the FTO substrate.



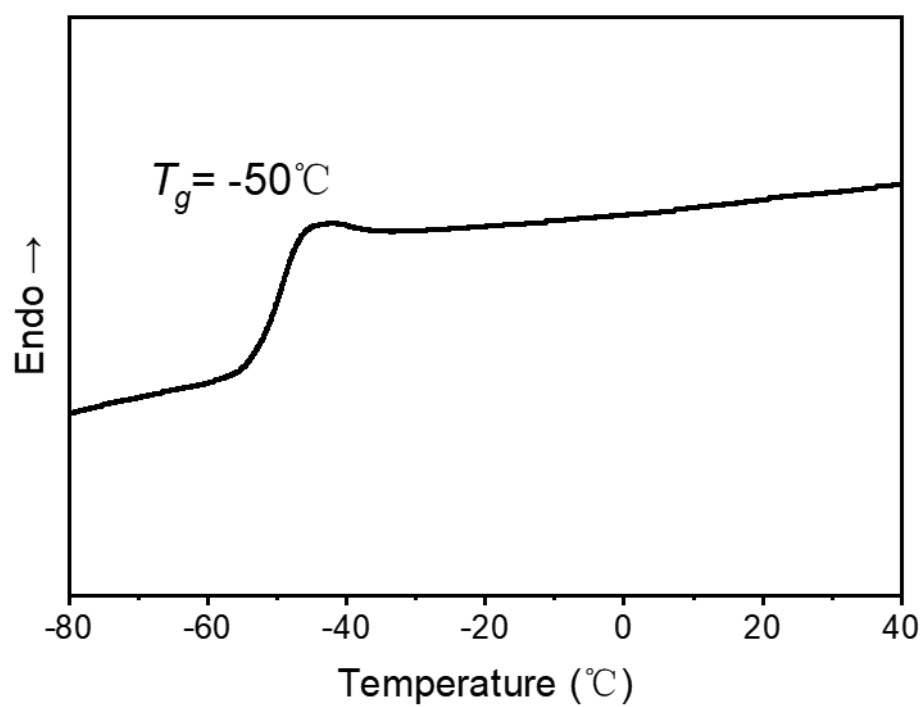
Supplementary Fig. 47. Hot-carrier temperatures as a function of the delay time for perovskite (SAM) film deposited on the FTO substrate.



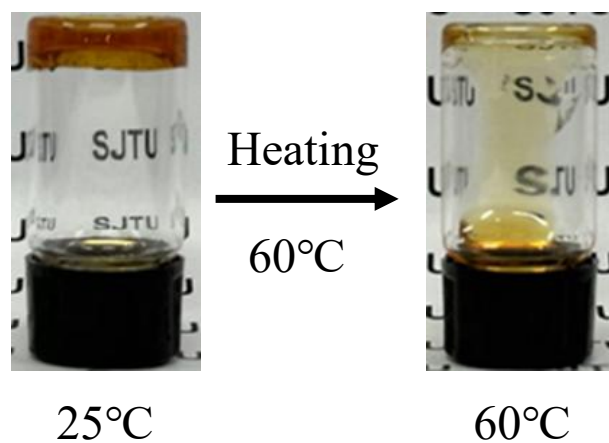
Supplementary Fig. 48. Pseudo color plot and of (a) perovskite (Me-4PACz) film and (b) perovskite (PhBr-4PACz) film detached from the substrate.



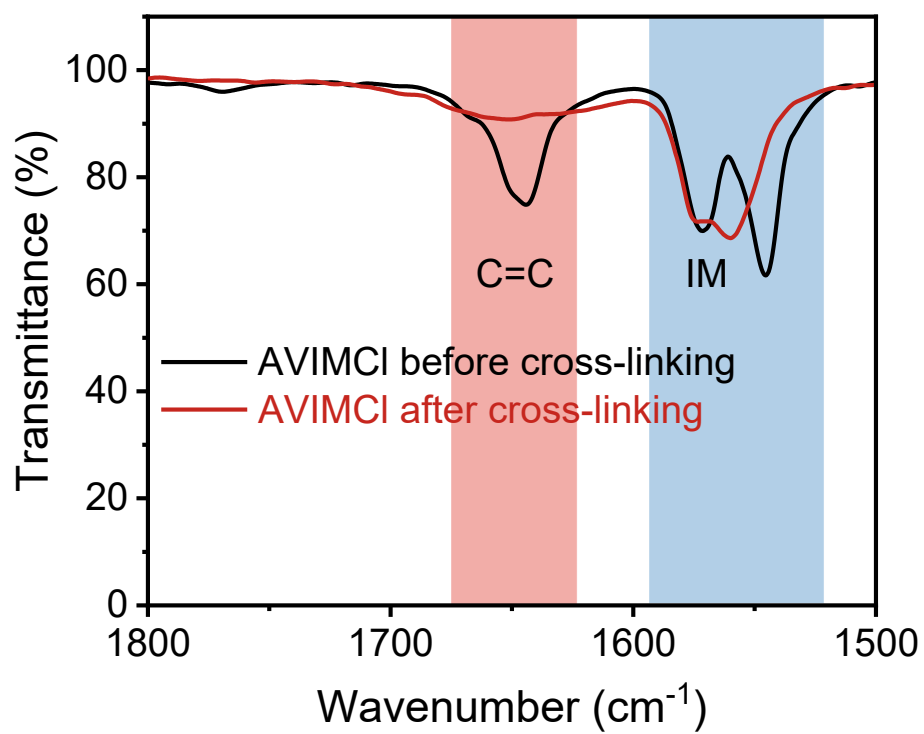
Supplementary Fig. 49. Hot-carrier temperatures as a function of the delay time for perovskite (SAM) film detached from the substrate.



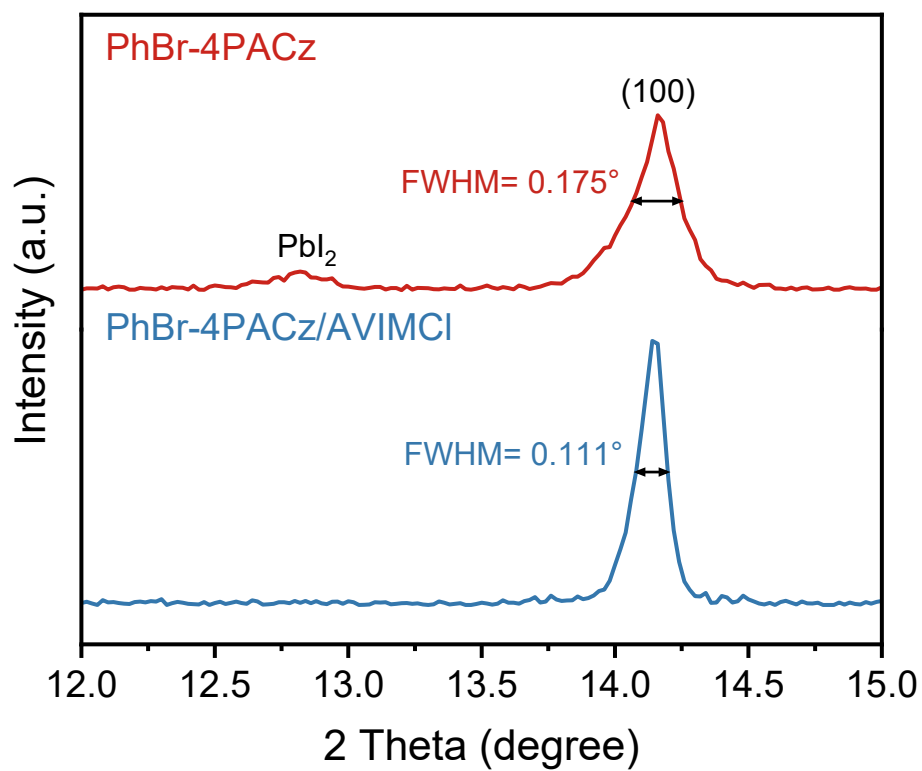
Supplementary Fig. 50. The glass transition temperature of AVIMCl.



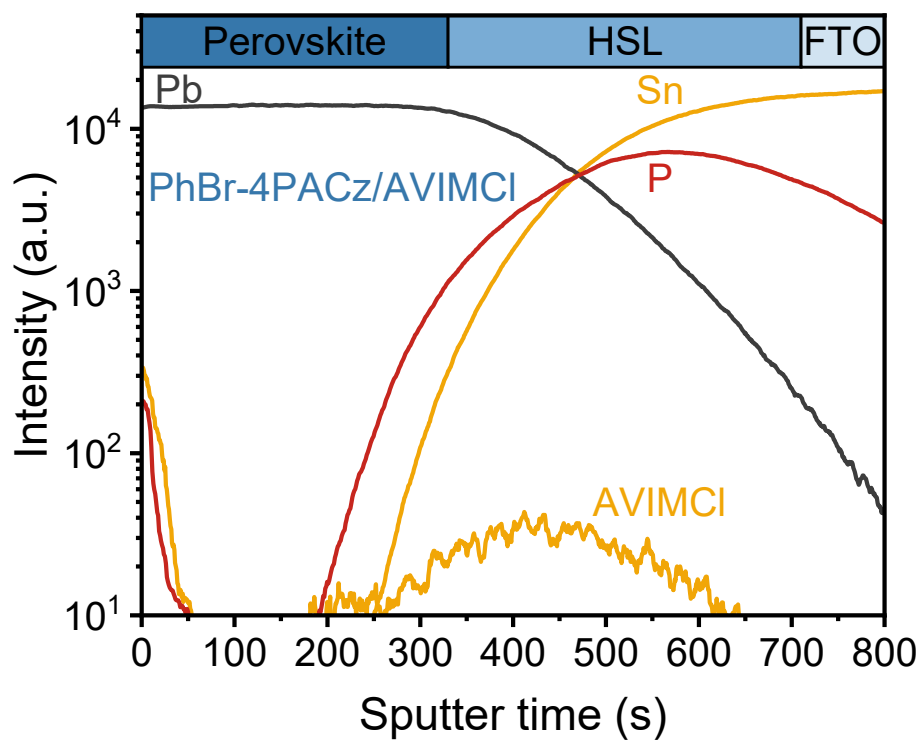
Supplementary Fig. 51. Photographs of AVIMCl at different temperatures, showing a liquefied process under heating.



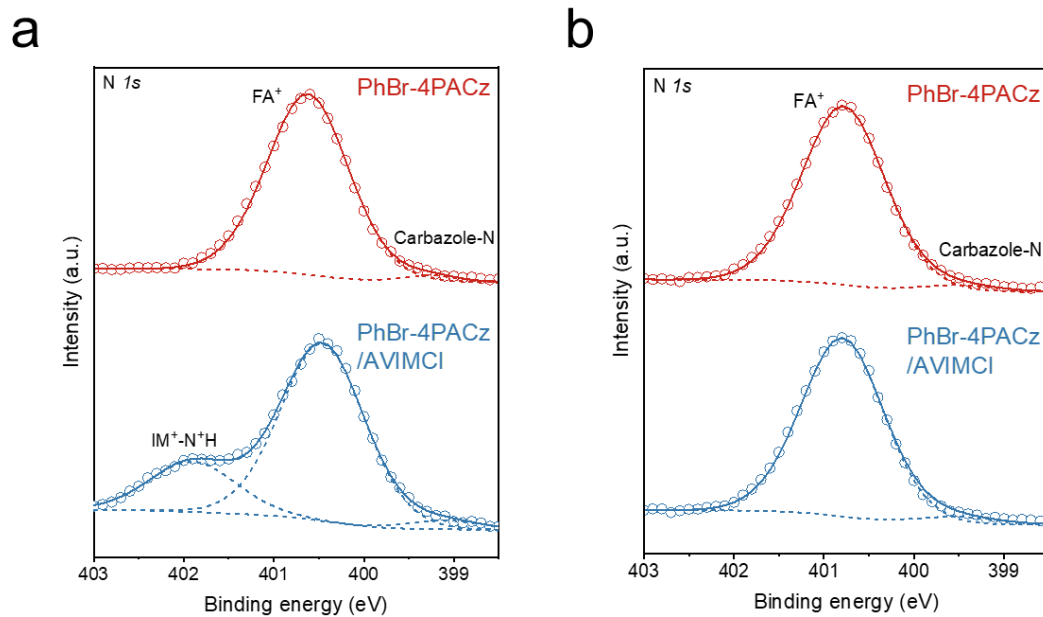
Supplementary Fig. 52. FTIR of AVIMCl before and after cross-linking under initiators.



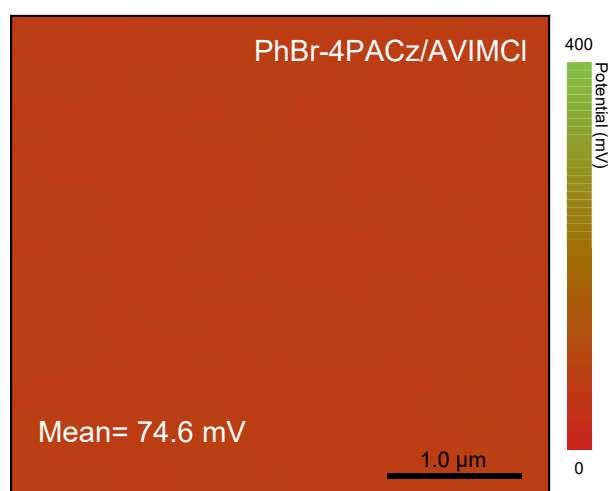
Supplementary Fig. 53. The zoomed-in XRD patterns of perovskite (PhBr-4PACz) films with and without AVIMCl treatment.



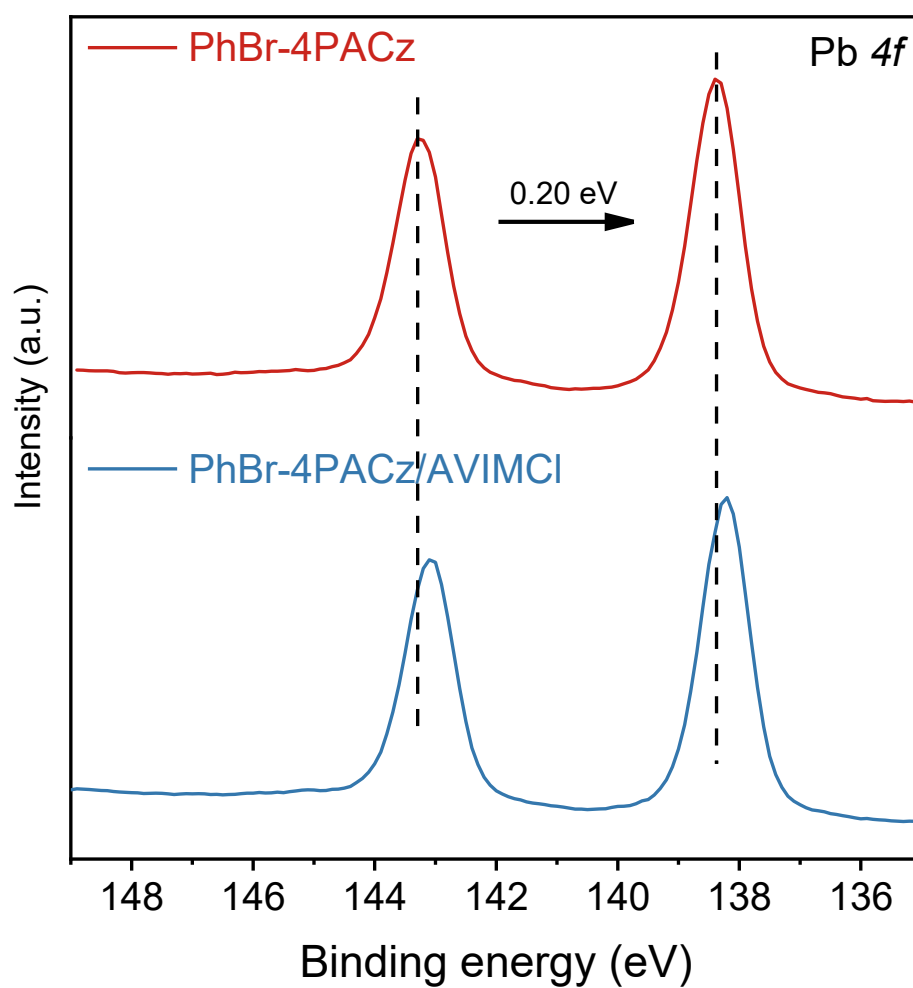
Supplementary Fig. 54. The ToF-SIMS profiles of unaged FTO/perovskite (PhBr-4PACz)/AVIMCI samples.



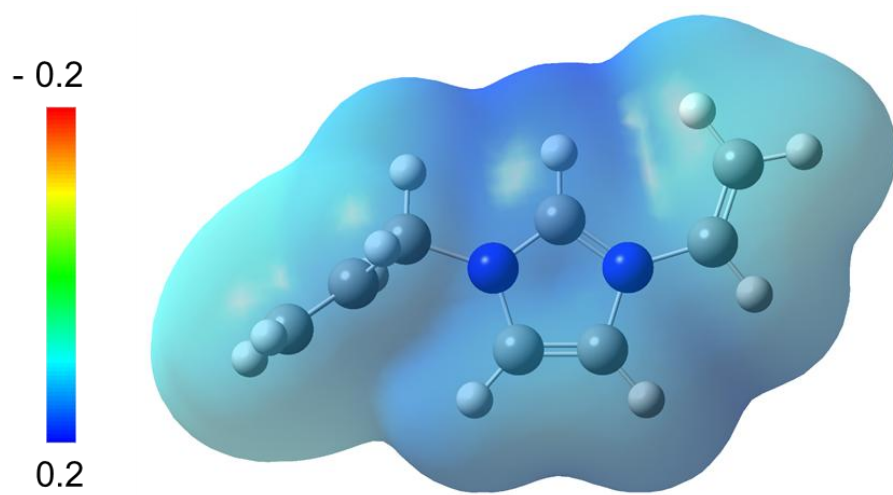
Supplementary Fig. 55. XPS spectra of N 1s for (a) top surface and (b) bottom surface of perovskite (PhBr-4PACz) films with or without AVIMCl treatment.



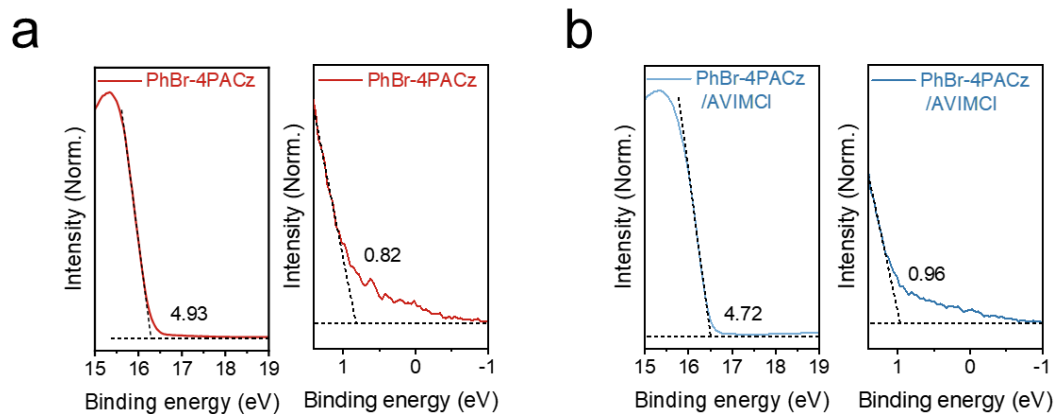
Supplementary Fig. 56. KPFM images of bottom surface of perovskite (PhBr-4PACz) film treated with AVIMCl.



Supplementary Fig. 57. XPS spectra of Pb 4f for top surface of perovskite (PhBr-4PACz) films with or without AVIMCl treatment.

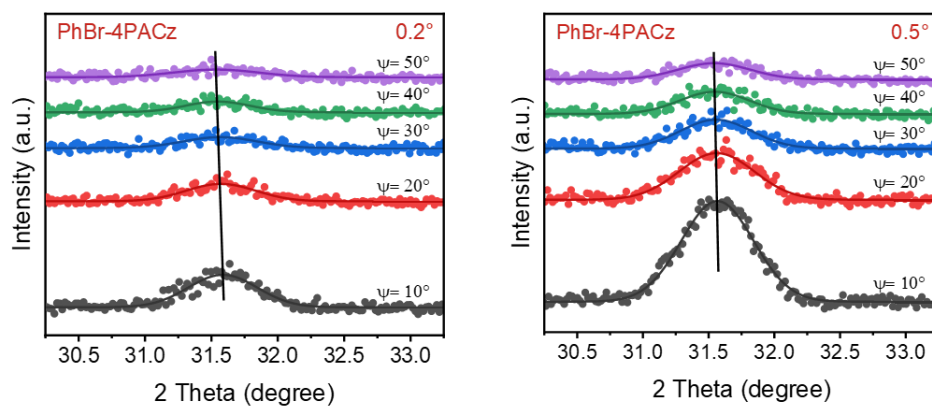


Supplementary Fig. 58. Electrostatic potential surfaces (ESP) of AVIM⁺.

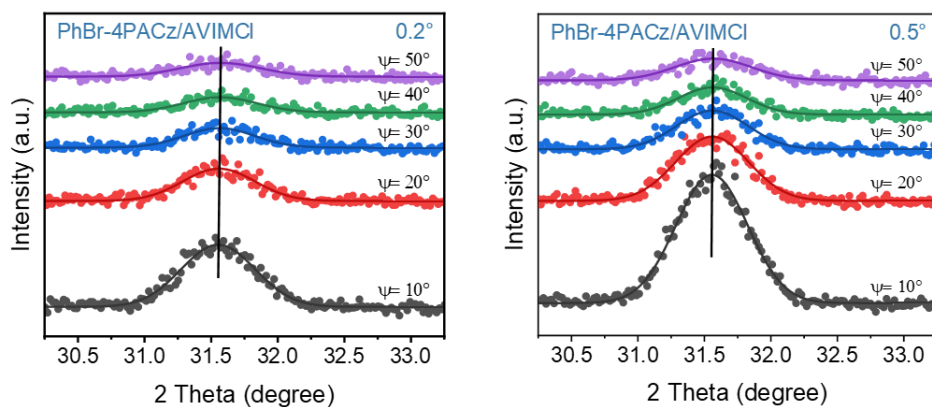


Supplementary Fig. 59. UPS spectra of secondary electron cutoff region and Fermi edge region for top surface of perovskite (PhBr-4PACz) film (a) without or (b) with AVIMCl treatment.

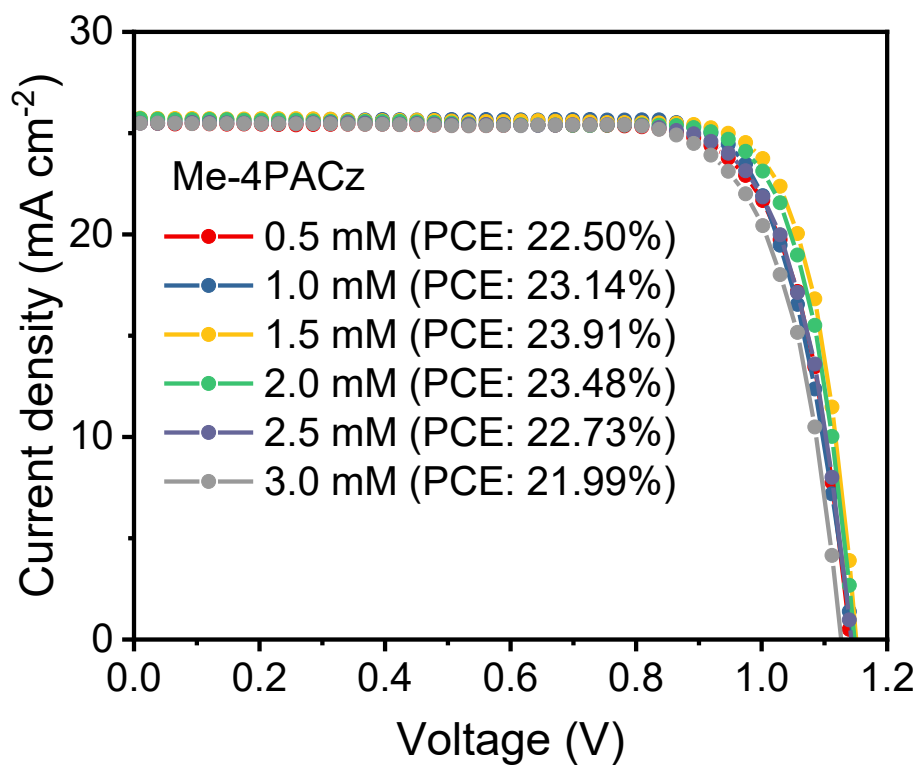
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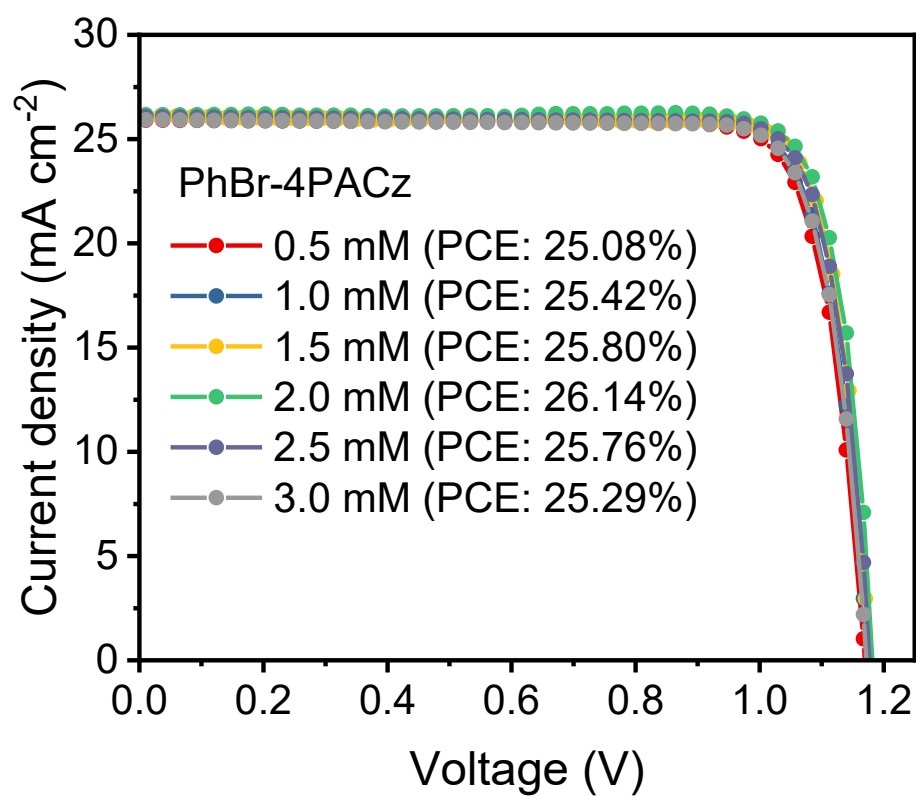
b



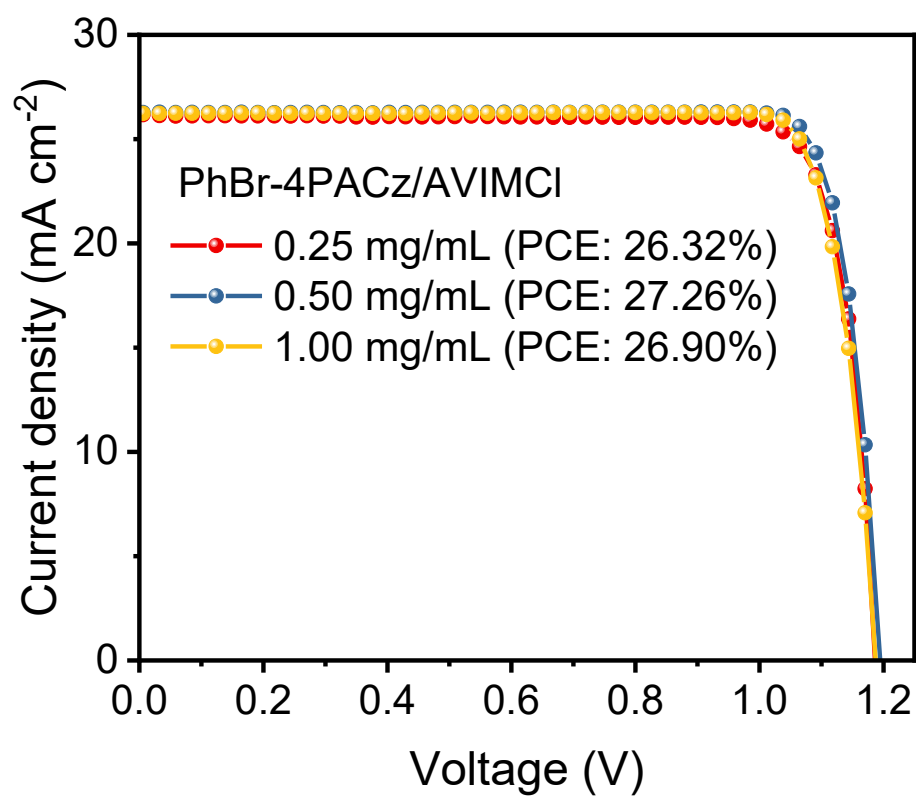
Supplementary Fig. 60. (a) GIXRD with different instrumental Ψ values of perovskite (PhBr-4PACz) film at different incident angles of 0.2° (left) and 0.5° (right). (b) GIXRD with different instrumental Ψ values of perovskite (PhBr-4PACz)/AVIMCl film at different incident angles of 0.2° (left) and 0.5° (right).



Supplementary Fig. 61. *J-V* curves of devices doped with different concentrations of Me-4PACz.



Supplementary Fig. 62. J - V curves of devices doped with different concentrations of PhBr-4PACz.



Supplementary Fig. 63. *J-V* curves of devices doped with different concentrations of AVIMCl.



福建省计量科学研究院
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(国家光伏产业计量测试中心)
National PV Industry Measurement and Testing Center



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国家认可
标志
注册号
CNAS L6731

检 测 报 告

Test Report

报告编号: 24Q3-00831

Report No.

客 户 信 息

Name of Customer

Shanghai Jiao Tong University and CATL

联 络 信 息

Contact Information

800 Dongchuan Road, Minhang District, Shanghai, China

物 品 名 称

Name of Items

Inverted Perovskite Solar Cell (IV)

型 号 / 规 格

Type / Specification

Inverted PSC

物 品 编 号

Items No.

A4

制 造 厂 商

Manufacturer

Chun-Chao Chen's group

物 品 接 收 日 期

Items Receipt Date

2024-11-26

检 测 日 期

Test Date

2024-11-27



批 准 人:

Approved by

黎健生

黎健生

核 验 员:

Checked by

何翔

何翔

检 测 员:

Test by

陈彩云

陈彩云

发布日期:

Date of Report

2024 年 12 月 02 日

Year

Month

Day



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本院/本中心地址: 福州市屏东路9-3号

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Supplementary Fig. 64a. Independent efficiency certification of one PhBr-4PACz/AVIMCl-based inverted device by an accredited institute of National PV Industry Measurement and Testing Center (Fujian Metrology Institute, China). The aperture area is 0.0737 cm² measured by NPVM.



福建省计量科学研究院
FUJIAN METROLOGY INSTITUTE
(国家光伏产业计量测试中心)
National PV Industry Measurement and Testing Center

报告编号: 24Q3-00831

Report No.

1. 检测机构说明:

Testing institutions that

本院为国家法定计量检定机构, 国家光伏产业计量测试中心依托本院检测技术开展检测。本院/本中心质量管理体系符合GB/T 27025 (ISO/IEC 17025, IDT) 标准要求。

The institute is a national legal metrological institution. National PV Industry Measurement and Testing Center carries out testing relying on the institute's testing technology. The Center's quality management system meets the requirements of GB/T 27025 (ISO/IEC 17025, IDT) standard.

2. 本次检测所依据的检测方法(代号及名称):

Reference documents for the test (code, name)

IEC 60904-1-2020 光伏器件-第一部分: 光伏电流-电压特性的测量; SJ/T 11630-2016 太阳能电池用硅片几何尺寸测试方法

3. 本次检测所使用的主要测量仪器:

Measurement standards used in this test

仪器名称 Name	仪器编号 Number	测量范围 Measuring Range	不确定度/或准确度等级/或最大允许误差 Uncertainty or Accuracy Class or Maximum Permissible Error	溯源机构名称/证书编号 Name of traceability institution/Certificate No.	有效期限 Due date
源表	10807C00878-2	电流: $10 \mu\text{A} \sim 1\text{A}$; 电压: $20\text{mV} \sim 20\text{V}$	测量: $\text{DCV}: U_{\text{rel}}=0.05\%, k=2$; DCI: $U_{\text{rel}}=0.05\%, k=2$ 输出: $\text{DCV}: U_{\text{rel}}=0.05\%, k=2$; DCI: $U_{\text{rel}}=0.05\%, k=2$	福建计量院24D2-01917	2025-04-17
太阳模拟器	2014-017	$(300 \sim 1200) \text{ nm}$; $(300 \sim 1200) \text{ W/m}^2$	光谱匹配度 $(300 \sim 360) \text{ nm}$: $U_{\text{rel}}=8.0\%$ ($k=2$); $(360 \sim 1200) \text{ nm}$: $U_{\text{rel}}=6.2\%$ ($k=2$); 辐照度比(辐照度不均匀度、辐照度时间不稳定性): $U_{\text{rel}}=1.2\%$ ($k=2$)	福建计量院24Q2-00981	2025-06-16
WPVS单晶硅标准电池	015-2014	$(300 \sim 1200) \text{ nm}$	$U_{\text{rel}}=1.3\%$ ($k=2$)	中国计量院 GXgf2023-01245	2025-03-29
自动影像测量仪	11656	X轴: $(0 \sim 195) \text{ mm}$; Y轴: $(0 \sim 195) \text{ mm}$	$U=2 \mu\text{m}$ ($k=2$)	福建计量院24A2-09430	2025-07-23
数字温度计	15-B	$(15 \sim 65) ^\circ\text{C}$	$U=0.1 ^\circ\text{C}$ ($k=2$)	福建计量院24B2-05374	2025-06-19

4. 检测地点及环境条件

Location and environmental condition for the test

地点: Room 108, Building 4, MinHou Scientific Research Base

Location

温度: $24.8 ^\circ\text{C}$ 相对湿度: 37 % 其它: /

Temperature

Relative Humidity

Others

5. 备注: /

Note

本报告提供的结果仅对本次被检的物品有效。

The data are valid only for the instrument(s) under testing.

检测报告续页专用

Continued page of test report

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Supplementary Fig. 64b. Independent efficiency certification of one PhBr-4PACz/AVIMCl-based inverted device by an accredited institute of National PV Industry Measurement and Testing Center (Fujian Metrology Institute, China). The aperture area is 0.0737 cm^2 measured by NPVM.



检测结果/说明:

Results of Test and additional explanation.

1. Standard Test Condition (STC): Total Irradiance: 1000 W/m²
Temperature: 25.0 °C
Spectral Distribution: AM1.5G

2.Measurement Data and I-V/P-V Curves under STC

Forward Scan

I_{sc} (mA)	V_{oc} (V)	I_{MPP} (mA)	V_{MPP} (V)	P_{MPP} (mW)	FF (%)	A (cm ²)
1.936	1.184	1.853	1.056	1.957	85.38	0.0737

Reverse Scan

I_{sc} (mA)	V_{oc} (V)	I_{MPP} (mA)	V_{MPP} (V)	P_{MPP} (mW)	FF (%)	A (cm ²)
1.936	1.186	1.863	1.069	1.992	86.76	0.0737

Mismatch factor: 1.019

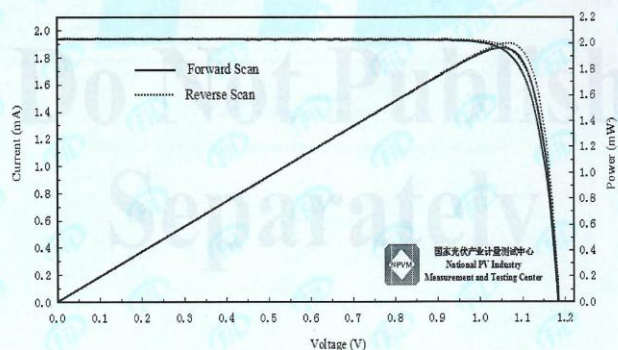


Figure 1. I-V and P-V characteristic curves of the measured sample under STC

检测报告续页专用

Continued page of test report

Supplementary Fig. 64c. Independent efficiency certification of one PhBr-4PACz/AVIMCl-based inverted device ($J-V$ characterization). by an accredited institute of National PV Industry Measurement and Testing Center (Fujian Metrology Institute, China). The certified efficiency is 27.03% under reverse scan.



检测结果/说明:

Results of Test and additional explanation.

3.Measurement Data and Curves for MPPT under STC

η (%)	26.50
P_{MPP} (mW)	1.953
I_{MPP} (mA)	1.842
V_{MPP} (V)	1.060

Note: Measurement data for MPPT under STC in the above table was the mean value acquired during the final 30 seconds of the 300 seconds test

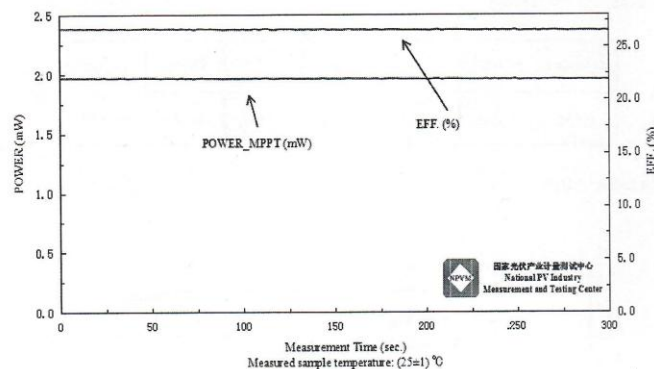
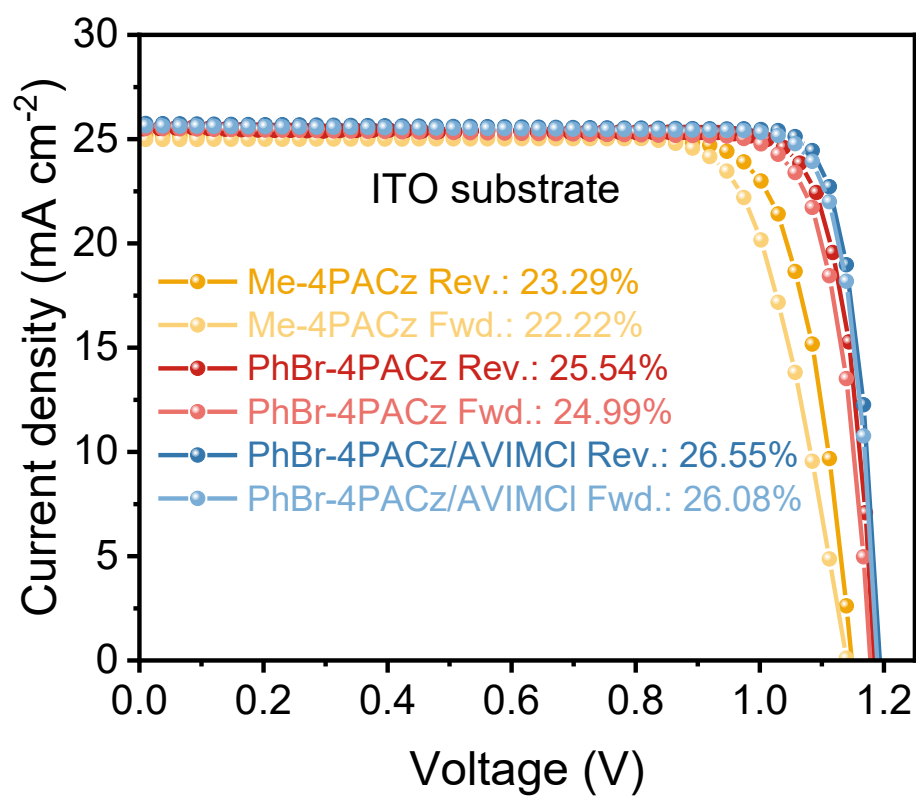


Figure 2. Measurement curves of the measured sample for MPPT

Supplementary Fig. 64d. Independent efficiency certification of one PhBr-4PACz/AVIMCl-based inverted device by an accredited institute of National PV Industry Measurement and Testing Center (Fujian Metrology Institute, China). The stabilized PCE of 26.50% is determined from MPPT for 300 s.



Supplementary Fig. 65. J - V curves of best-performing devices obtained on ITO substrate.

		福建省计量科学研究院 FUJIAN METROLOGY INSTITUTE (国家光伏产业计量测试中心) National PV Industry Measurement and Testing Center	
			
检测报告 Test Report			
报告编号: 24Q3-00048 Report No.			
客户名称 Name of Customer	Shanghai Jiao Tong University and CATL (SJTU/CATL)		
联络信息 Contact Information	800 Dongchuan Road, Minhang District, Shanghai, China		
物品名称 Name of Items	Flexible Inverted Perovskite Solar Cell (IV)		
型号/规格 Type /Specification	Flexible PSC		
物品编号 Items No	A23		
制造厂商 Manufacturer	Chun-Chao Chen' s group		
物品接收日期 Items Receipt Date	2024-01-22		
检测日期 Test Date	2024-01-22		
	批准人 Approved by	黎健生	黎健生
	核验员 Checked by	何翔	何翔
	检测员 Test by	陈彩云	陈彩云
发布日期 Date of Report		2024 年 01 月 26 日 Year month Day	 扫一扫 查真伪
本院/本中心地址: 福州市屏东路 9-3 号 Address : 9-3 Pingdong Road,Fuzhou,China 网址: www.fjli.net Web Site		电话: 0591-87845050 Telephone 咨询电话: 0591-87845050 Inquire line	传真: 0591-87808417 Fax 投诉电话: 0591-87823025 Complaint Tel
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Supplementary Fig. 66a. Certified performance of flexible inverted perovskite solar cell with an area of 7.56 mm² based on PhBr-4PACz/AVIMCl at NPVM.



福建省计量科学研究院
FUJIAN METROLOGY INSTITUTE
(国家光伏产业计量测试中心)
National PV Industry Measurement and Testing Center

报告编号: 24Q3-00048
Report No.

1. 检测机构说明:

Testing institutions that

本院为国家法定计量检定机构, 国家光伏产业计量测试中心依托本院检测技术开展检测。本院/本中心质量管理体系符合 GB/T 27025 (ISO/IEC 17025, IDT) 标准要求。

The institute is a national legal metrological institution. National PV Industry Measurement and Testing Center carries out testing relying on the institute's testing technology. The Center's quality management system meets the requirements of GB/T 27025 (ISO/IEC 17025, IDT) standard.

2. 本次检测所依据的检测方法(代号及名称):

Reference documents from the test (code, name)

IEC 60904-1-2020 光伏器件-第一部分: 光伏电流-电压特性的测量; SJ/T 11630-2016 太阳能电池用硅片几何尺寸测试方法

3. 本次检测所使用的主要测量仪器:

Measurement standards used in this test

仪器名称 Name	仪器编号 Number	测量范围 Measuring Range	不确定度/或准确度等级/或最大允许误差 Uncertainty or Accuracy Class or Maximum Permissible Error	溯源机构名称/证书编号 Name of traceability Institution/Certificate No.	有效期限 Due date
源表	10807C00878-2	电流: $-10\mu\text{A}$ ~ 1A ; 电压: 20mV ~ 20V	测量: $\text{DCV}; U_{\text{ref}}=0.05\%, k=2$; $\text{DCI}; U_{\text{ref}}=0.05\%, k=2$ 输出: $\text{DCV}; U_{\text{ref}}=0.05\%, k=2$; $\text{DCI}; U_{\text{ref}}=0.05\%, k=2$	福建计量院 23D2-02429	2024-04-16
太阳模拟器	2014-017	(300~1200) nm; (800~1200) W/m^2	光谱匹配度(300~360) nm: $U_{\text{ref}}=8.0\% (k=2)$; (360~1200) nm: $U_{\text{ref}}=6.2\% (k=2)$; 辐照度比(辐照度不均匀度、辐照度时间不稳定性): $U_{\text{ref}}=1.2\% (k=2)$	福建计量院 23Q2-00526	2024-06-13
WPVS 单晶硅标准电池	015-2014	(300~1200) nm	$U_{\text{ref}}=1.3\% (k=2)$	中国计量院 GXgf2023-01245	2025-03-29
自动影像测量仪	11656	X 轴: (0~195) mm; Y 轴: (0~195) mm	$U=2\mu\text{m} (k=2)$	福建计量院 23A2-11241	2024-07-13
/	/	/	/	/	/



4. 检测地点及环境条件:

Location and environmental condition for the test

地点: Room 108, Building 4, MinHou Scientific Research Base

Location

温度: $25.4\text{ }^{\circ}\text{C}$
Temperature

相对湿度: 36 %
Relative Humidity

其它: /
Others

5. 备注:

Note

本报告提供的结果仅对本次被检的物品有效。

The data are valid only for the instrument(s) under testing.

检测报告续页专用

Continued page of test report

第 2 页/共 6 页
Page of Pages

Supplementary Fig. 66b. Certified performance of flexible inverted perovskite solar cell with an area of 7.56 mm^2 based on PhBr-4PACz/AVIMCl at NPVM.



检测结果/说明:

Results of Test and Additional Explanation.

1 Standard Test Condition (STC): Total Irradiance: 1000 W/m²

Temperature: 25.0 °C

Spectral Distribution: AM1.5G

2 Measurement Data and I-V/P-V Curves under STC

Forward Scan

I_{sc} (mA)	V_{oc} (V)	I_{MPP} (mA)	V_{MPP} (V)	P_{MPP} (mW)	FF (%)	η (%)	A (cm ²)
1.856	1.181	1.771	1.033	1.829	83.44	24.19	0.07561

Reverse Scan

I_{sc} (mA)	V_{oc} (V)	I_{MPP} (mA)	V_{MPP} (V)	P_{MPP} (mW)	FF (%)	η (%)	A (cm ²)
1.857	1.181	1.777	1.042	1.852	84.45	24.49	0.07561

Mismatch Factor: 1.001

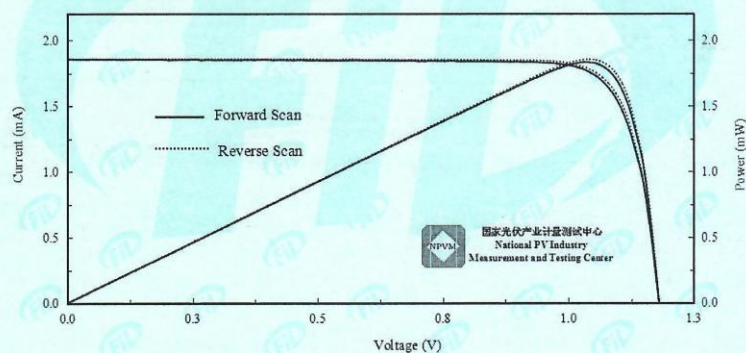


Figure 1. I-V and P-V characteristic curves of the measured sample under STC

Supplementary Fig. 66c. Certified performance of flexible inverted perovskite solar cell with an area of 7.56 mm² based on PhBr-4PACz/AVIMCl at NPVM. The certified efficiency is 24.49% under reverse scan.



检测结果/说明:

Results of Test and Additional Explanation.

3 Measurement Data and Curves for MPPT under STC

η (%)	24.41
P_{MPP} (mW)	1.846
I_{MPP} (mA)	1.794
V_{MPP} (V)	1.029

Note: Measurement data for MPPT under STC in the above table was the mean value acquired during the final 30 seconds of the 300 seconds test

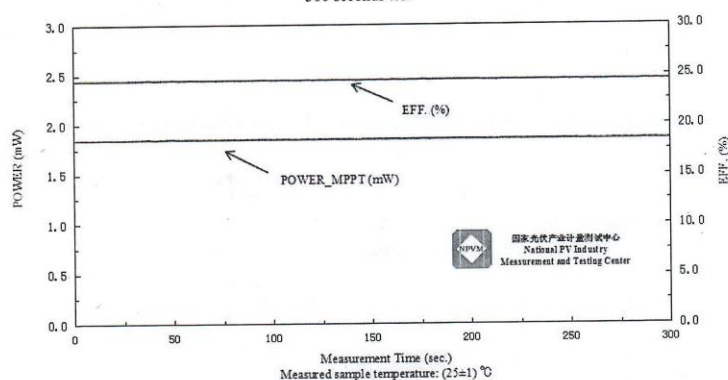
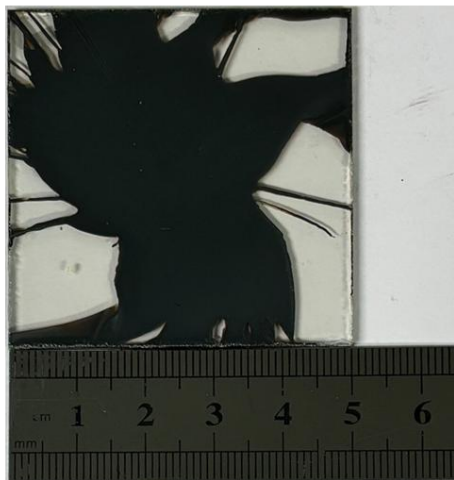


Figure 2. Measurement curves of the measured sample for MPPT

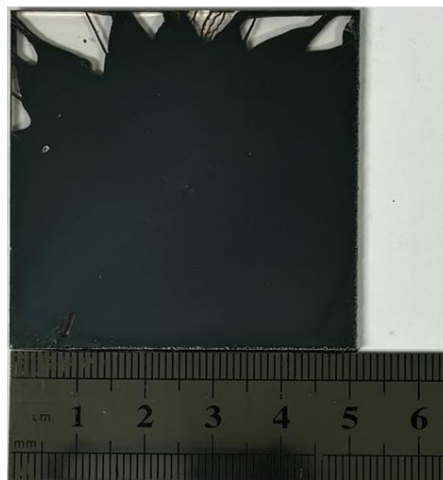
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Supplementary Fig. 66d. Certified performance of flexible inverted perovskite solar cell with an area of 7.56 mm² based on PhBr-4PACz/AVIMCl at NPVM. The stabilized PCE of 24.41% is determined from MPPT for 300 s.

a FTO/Me-4PACz/perovskite



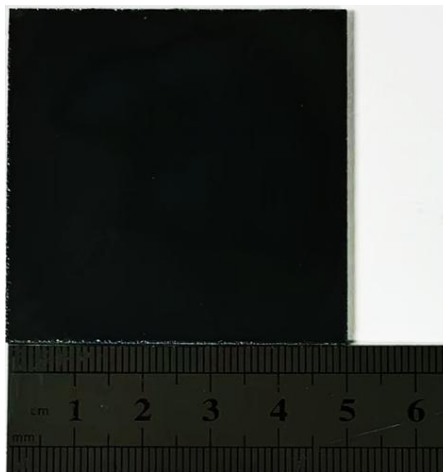
b FTO/PhBr-4PACz/perovskite



c FTO/perovskite (Me-4PACz)

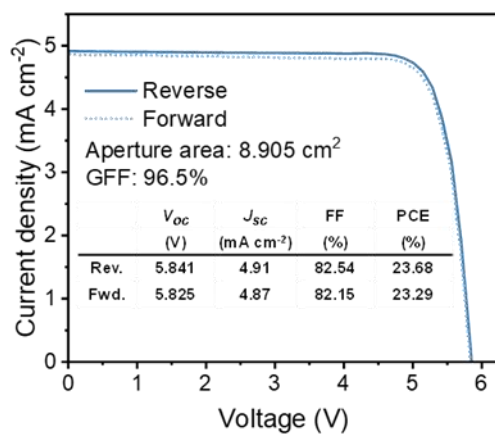


d FTO/perovskite (PhBr-4PACz)



Supplementary Fig. 67. Photographic images of a neat perovskite film on a $5 \times 5 \text{ cm}^2$ (a) FTO/Me-4PACz substrate and (b) FTO/PhBr-4PACz substrate. Photographic images of (c) a perovskite (Me-4PACz) film and (d) a perovskite (PhBr-4PACz) film on a $5 \times 5 \text{ cm}^2$ FTO substrate.

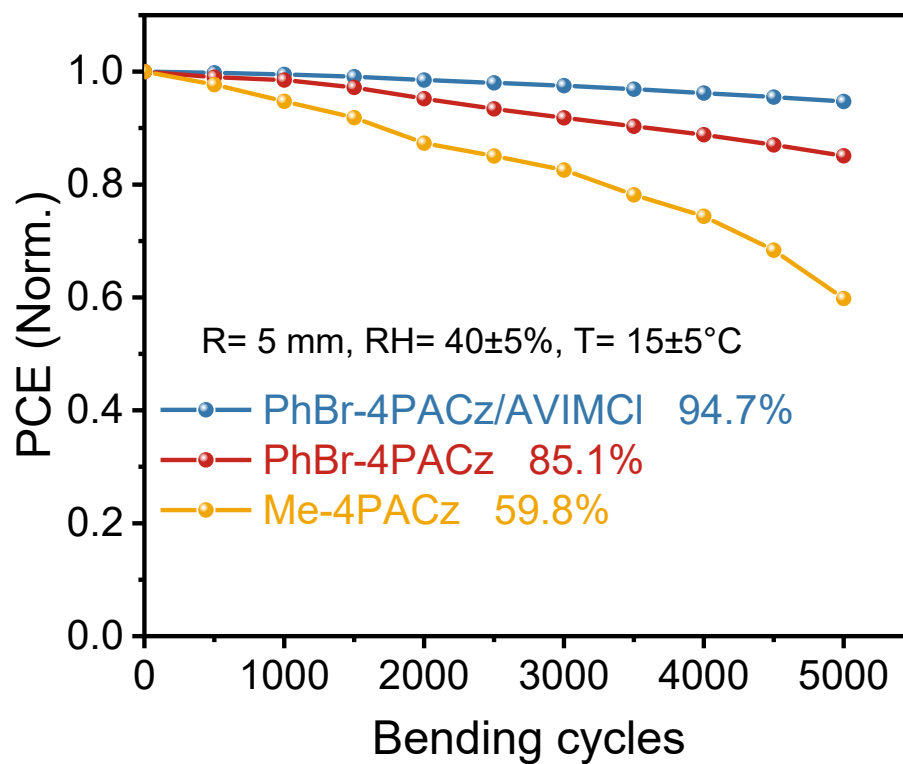
a



b



Supplementary Fig. 68. (a) J - V curves of the champion PhBr-4PACz/AVIMCl-based mini-module with an aperture area of 8.905 cm^2 on a $5 \times 5 \text{ cm}^2$ substrate. (b) Photographic image of mini-module with an aperture area of 8.905 cm^2 on a $5 \times 5 \text{ cm}^2$ substrate.



Supplementary Fig. 69. 5000 bending cycles at a radius of 5 mm on three flexible devices under environmental conditions ($RH = 40 \pm 5\%$, $T = 15 \pm 5^\circ\text{C}$).

Supplementary Table 1. Area of N *1s* and Sn *3d* for XPS spectra and related N/Sn ratios of FTO/SAM substrates after washing off the perovskite (SAM) films.

Sample	Area				
	N <i>1s</i>	P <i>2p</i>	Sn <i>3d</i>	N/Sn (%)	P/Sn (%)
Me-4PACz	3441	808	591174	0.58	0.14
PhBr-4PACz	3233	903	478327	0.68	0.19

Supplementary Table 2. Area of O *1s* and Pb *4f* for XPS spectra and related O/Pb ratios of bottom surface of perovskite (SAM) films.

Sample	Area		
	O <i>1s</i>	Pb <i>4f</i>	O/Pb (%)
Me-4PACz	9895	285948	3.46
PhBr-4PACz	16417	235043	6.98

Supplementary Table 3. PLQY, QFLS and $\Delta V_{\text{non-rad}}$ results of perovskite (Me-4PACz), perovskite (PhBr-4PACz) and AVIMCl modified perovskite (PhBr-4PACz) films, respectively.

Sample	PLQY (%)	QFLS (eV)	$\Delta V_{\text{non-rad}}$ (mV)
Me-4PACz	1.4	1.17	110
PhBr-4PACz	4.3	1.20	81
PhBr-4PACz/AVIMCl	5.2	1.21	75

Supplementary Table 4. The PL decay fitting parameters of perovskite (SAM) films, excited from the glass side.

Sample	A_1	τ_1 (ns)	A_2	τ_2 (ns)	τ_{ave} (ns)
Me-4PACz	0.959	34.6	0.041	590.8	270
PhBr-4PACz	0.973	25.1	0.027	1600.1	1033

Supplementary Table 5. Device performance parameters at different SAM doping concentrations.

Sample	Concentration (mmol mL ⁻¹)	V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF (%)	PCE (%)
Me-4PACz	0.5	1.141	25.50	77.33	22.50
	1.0	1.146	25.66	78.69	23.14
	1.5	1.151	25.75	80.67	23.91
	2.0	1.148	25.73	79.49	23.48
	2.5	1.143	25.51	77.95	22.73
	3.0	1.127	25.49	76.55	21.99
PhBr-4PACz	0.5	1.170	25.91	82.73	25.08
	1.0	1.174	26.00	83.28	25.42
	1.5	1.176	26.05	84.21	25.80
	2.0	1.182	26.17	84.51	26.14
	2.5	1.177	26.08	83.91	25.76
	3.0	1.172	25.95	83.15	25.29

Supplementary Table 6. Performance parameters of the best performing devices with Me-4PACz, PhBr-4PACz, and PhBr-4PACz/AVIMCl.

Sample		V_{OC} (V)	J_{SC} (mA cm ⁻²)	FF (%)	PCE (%)
Me-4PACz	Forward scan	1.144	25.62	78.88	23.12
	Reverse scan	1.151	25.75	80.67	23.91
PhBr-4PACz	Forward scan	1.176	26.10	83.53	25.64
	Reverse scan	1.182	26.17	84.51	26.14
PhBr-4PACz /AVIMCl	Forward scan	1.189	26.25	86.04	26.85
	Reverse scan	1.193	26.28	86.95	27.26

Supplementary Table 7. Device performance parameters of PhBr-4PACz/AVIMCl at different AVIMCl concentrations.

Sample	AVIMCl concentration (mg mL ⁻¹)	V_{OC} (V)	J_{SC} (mA cm ⁻²)	FF (%)	PCE (%)
PhBr-4PACz /AVIMCl	0.25	1.187	26.18	84.70	26.32
	0.50	1.193	26.28	86.95	27.26
	1.00	1.188	26.22	86.36	26.90

Supplementary Table 8. Performance comparison of recently reported inverted PSCs fabricated using conventional sequential deposition processes.

Reference	V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF (%)	PCE (%)	Certified PCE (%)
<i>Nature</i> 628, 93-98 (2024). ¹	1.19	24.94	85.90	25.6	25.0
<i>Science</i> 383, 524-531 (2024). ²	1.19	25.0	86.0	25.6	25.3
<i>Nat. Energy</i> 9, 308 (2024). ³	1.17	25.67	85.3	25.7	25.06
<i>Nature</i> 632, 301-306 (2024). ⁴	1.18	26.0	85.1	26.1	25.7
<i>Nature</i> 635, 874-881 (2024). ⁵	1.17	26.3	84.5	26.1	25.85
<i>Science</i> 386, 1256-1264 (2024). ⁶	1.195	26.0	84	26.2	25.90
<i>Nature</i> 632, 536-542 (2024). ⁷	1.201	26.30	84.50	26.69	26.54
<i>Science</i> 386, 898-902 (2024). ⁸	1.18	26.5	85.5	26.7	26.28
<i>Science</i> 384, 189-193 (2024). ⁹	1.18	26.4	86.2	26.9	26.15
<i>Nat. Energy</i> 10, 181-190 (2025). ¹⁰	1.173	25.61	85.1	25.57	25.40
<i>Nat. Energy</i> 10, 342-353 (2025). ¹¹	1.174	26.2	85.5	26.3	26.08
<i>Nat. Photonics</i> 19, 258-263 (2025). ¹²	1.175	25.87	84.88	25.80	25.70

Supplementary Table 9. Performance comparison of recently reported inverted PSCs fabricated using co-deposition processes.

Reference	V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF (%)	PCE (%)	Certified PCE (%)
<i>Nat Energy</i> 8, 462–472 (2023). ¹³	1.19	24.78	83.1	24.20	/
<i>Mater. Horiz.</i> 10, 2609–2617 (2023). ¹⁴	1.13	24.4	80	22.2	/
<i>Nature</i> 620, 545–551 (2023). ¹⁵	1.187	25.69	84.73	25.86	25.39
<i>Angew. Chem. Int. Ed.</i> 64, e202424910 (2025). ¹⁶	1.18	25.95	85.07	26.05	25.49
<i>Joule</i> 9, 101880 (2025). ¹⁷	1.168	26.19	86.3	26.40	25.92
<i>Nat. Photon.</i> (2025) doi.org/10.1038/s41566-025-01725-x. ¹⁸	1.200	26.11	85.63	26.83	26.72
This work	1.193	26.28	86.95	27.26	27.03

Supplementary Table 10. P-i-n device performance parameters obtained on ITO substrate.

Sample		V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF (%)	PCE (%)
Me-4PACz	Forward scan	1.140	25.02	77.83	22.22
	Reverse scan	1.148	25.15	80.67	23.29
PhBr-4PACz	Forward scan	1.178	25.43	83.13	24.99
	Reverse scan	1.184	25.50	84.59	25.54
PhBr-4PACz /AVIMCl	Forward scan	1.188	25.62	85.68	26.08
	Reverse scan	1.194	25.69	86.56	26.55

Supplementary Note 1: Quasi-Fermi level splitting (QFLS) and non-radiative recombination voltage loss ($\Delta V_{\text{non-rad}}$) calculating methods

Radiative QFLS ($QFLS_{\text{rad}}$) are obtained from the following equation:

$$QFLS_{\text{rad}} = K_B T \ln \left(\frac{J_G}{J_{\text{rad},0}} \right)$$

where K_B , T , J_G and $J_{\text{rad},0}$ are Boltzmann constant, thermal equilibrium temperature, generation current density, dark radiative saturation current density, respectively. J_G is the integral product of the EQE spectrum, and $J_{\text{rad},0}$ can be determined by the following equation:^{19,20}

$$J_{\text{rad},0} = q \int EQE_{\text{PV}}(E) \phi_{\text{bb}}(E) dE$$

where q is the elementary charge, E is the photo energy, $\phi_{\text{bb}}(E)$ is the black body photon flux, which can be determined by the following equation:

$$\phi_{\text{bb}}(E) = \frac{2\pi E^2}{(h^3 c^2)} \exp \left(\frac{-E}{k_B T} \right)$$

Where h the Planck constant and c is the light velocity in vacuum.

Internal QFLS can be calculated based on the relationship with PLQY according to following equation:

$$QFLS = QFLS_{\text{rad}} + k_B T \ln(PLQY)$$

According to the Shockley-Queisser (S-Q) limit, the total loss of Voltage is divided into two parts:²¹

$$E_g = qV_{\text{OC}} + q\Delta V_{\text{total}} = qV_{\text{OC}} + q\Delta V_1 + q\Delta V_2$$

where q is the elementary charge, $q\Delta V_1$ is thermodynamic and radiative recombination loss, which can be determined by the following equation:

$$q\Delta V_1 = E_g - qV_{\text{rad}} = E_g - QFLS_{\text{rad}}$$

$q\Delta V_2$ ($\Delta V_{\text{non-rad}}$) is non-radiative recombination loss, which can be determined by the following equation:

$$q\Delta V_2 = E_g - q\Delta V_1 - QFLS$$

Supplementary Note 2: TRPL analysis

The TRPL data were fitted using a biexponential:²²

$$I(t) = A_1 \exp(-t/\tau_1) + A_2 \exp(-t/\tau_2)$$

Here, τ_1 is the fast transient component, τ_2 is the slow component, A_1 and A_2 are constants. The average lifetime τ_{avg} was calculated using the following formula:

$$\tau_{\text{avg}} = \frac{\sum A_i \tau_i^2}{\sum A_i \tau_i}$$

Supplementary Note 3: Analysis D based on confocal microscopy

To characterize the diffusion, we apply a Gaussian fit to the PL profiles at different time delays. This allows us to extract the standard deviation $\sigma_x(t)$ that can be interpreted as the instantaneous diffusion length at time t (see Supplementary Fig. 30). In Fig. 2L, we show these standard deviations as a function of time after excitation obtained from the Gaussian fits. In a classical diffusive scenario, the quantity $\sigma_x(t)$ follows the equation:²³

$$\sigma_x^2(t) = \sigma_x^2(0) + 2Dt$$

where D is the carrier diffusion coefficient.

Supplementary Note 4: Calculation of hot carrier temperature

The carrier temperature (T_c) can thus be obtained by fitting the high energy tail using the Maxwell-Boltzmann distribution function:²⁴

$$\frac{\Delta A}{A} \propto e^{-\frac{(E-E_f)}{k_B T_c}}$$

To ensure that the hot carriers have reached a quasi-equilibrium via the carrier-carrier scattering after the initial fs-pulse excitation, the hot carrier temperatures are analyzed from TA spectra after 0.4 ps time delay.

Supplementary Note 5: Estimation of hot-carrier diffusion length

The hot-carrier diffusion length L_{hot} can be estimated by:²⁵

$$L_{hot} = \sqrt{D_{hot} \times \tau_{hot}}$$
$$D_{hot} = \frac{\mu \times k_B \times T_c}{e}$$

Where, D_{hot} and τ_{hot} are diffusion coefficient and lifetime of hot carrier, respectively. μ is the carrier mobility, and e is the electronic charge (1.602×10^{-19} J V⁻¹). Here, the carrier mobility is taken as ~ 17 cm² V⁻¹ s⁻¹ at a fluence of approximately 2.8 μ J/cm², according to the previous report.²⁶

Supplementary Note 6: Transient reflectance spectroscopy analysis

The measured surface carrier kinetics can be modelled with diffusion and surface recombination/transfer model. The carriers are generated with profile:^{27,28}

$$N(x, 0) = N_0 e^{-ax}$$

where N_0 is the initial surface carrier density. If the traces are normalized, then N_0 is equal to 1. The values for the absorption coefficient at different pump-photon energies for samples are known. The carrier movement obeys the diffusion and surface recombination/transfer equation:

$$\begin{aligned}\frac{\partial N(x, t)}{\partial t} &= D \frac{\partial^2 N(x, t)}{\partial x^2} - \frac{N(x, t)}{\tau_B} \\ \left. \frac{\partial N(x \cdot t)}{\partial t} \right|_{x=0} &= \frac{S_F}{D} N(0, t) \\ \left. \frac{\partial N(x \cdot t)}{\partial t} \right|_{x=L} &= -\frac{S_L}{D} N(L, t)\end{aligned}$$

where $N(x, t)$ is the carrier density as a function of depth (x) and time (t), D is the ambipolar diffusion coefficient and τ_B is the bulk carrier lifetime. Generally, in perovskite material, τ_B is on the order of hundreds ns, in the 1200 ps probe window, it can be neglected. The boundary condition can be described by the followings, for the front surface, surface recombination/transfer will dominate (S_F). For the back surface, surface recombination/transfer happens with S_L . For pure perovskites, the back surface and front surface can be assumed to have the same surface recombination velocity. For samples with HSL, the front surface transfer velocity is different from the back where the back (S_L) takes the value of pure perovskites. L is the thickness of the film.

Supplementary Note 7: GIXRD analysis

According to Bragg's law and generalized Hooke's law, the residual strain can be calculated as the slope of the following function:²⁹⁻³¹

$$\sigma = -\frac{E\pi}{2(1+\nu)180^\circ} \cot\theta_0 \frac{\partial(2\theta)}{\partial \sin^2 \varphi}$$

where φ , E , 2θ , θ_0 , and ν represent the angle of diffraction vector corresponding to the sample surface normal, perovskite modulus, scattering angle of the actual perovskite, half of the scattering angle ($2\theta_0$) corresponding to a given diffraction peak for stress-free perovskite, and Poisson's ratio of the perovskite, respectively.

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